

CASSETTE

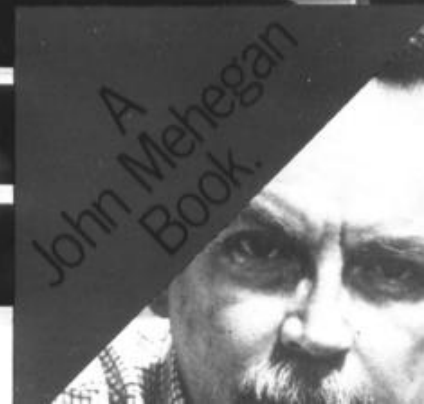
IMPROVISING Jazz Piano.

by John Mehegan

A brilliant and instructive treatise on jazz piano improvisation, by one of the most distinguished writers on jazz technique. Deals with all aspects, including a survey of piano styles from 1900 to the present day.



A
John Mehegan
Book.



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Chapter 1

The Harmonic Vocabulary (Introduction)

The principle of jazz improvisation involves the abandoning of the melody and the creating of new ideas from the resources of the harmony (chord changes) of a tune, arpeggios, modes, ornamental tones, etc. Because of this, it is essential that the student have a clear and concise idea of the use of harmony in dealing with jazz improvisation.

The diatonic system dating back to the 17th Century forms the basis for the jazz harmonic vocabulary. The diatonic system divides into two sub - systems, the triad system and the seventh chord system.

The Triad System:

The student is advised to be very familiar with the twelve major scales and, at least, the twelve harmonic minor scales before proceeding with this text.

Lesson: Play the scale - tone triads of Fig. 1 through Fig. 13 in both hands, with the left hand in the octave immediately below the right hand, until they can be played automatically from memory in any key.

Fig. 1 Key of C

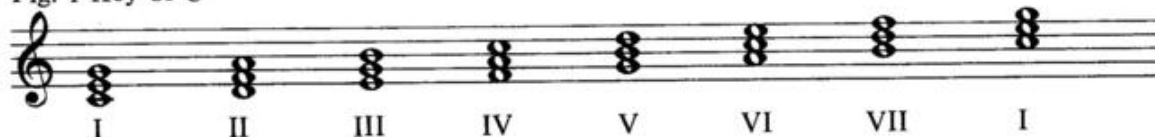


Fig. 2 Key of G

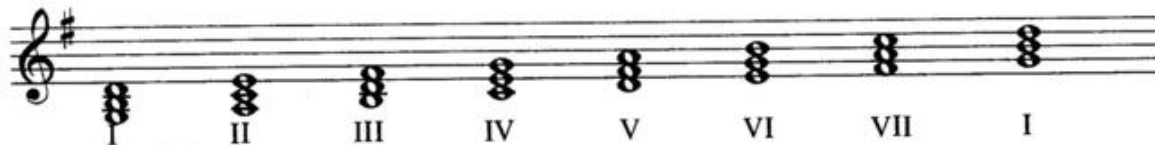


Fig. 3 Key of D

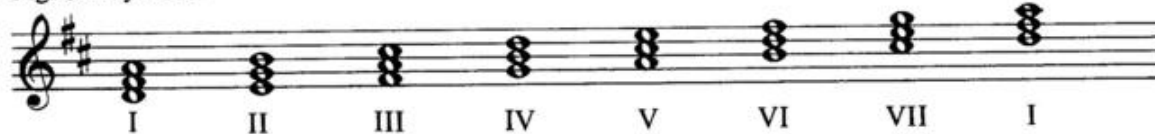


Fig. 4 Key of A

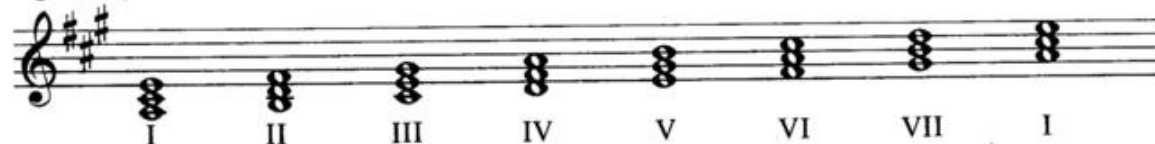


Fig. 5 Key of E



Fig. 6 Key of B

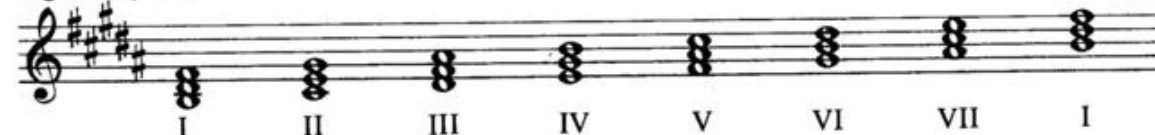


Fig. 7 Key of F $\sharp$ (Enharmonic of G $\flat$)

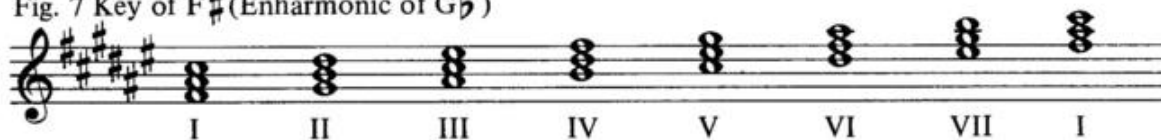


Fig. 8 Key of G $\flat$ (Enharmonic of F $\sharp$)

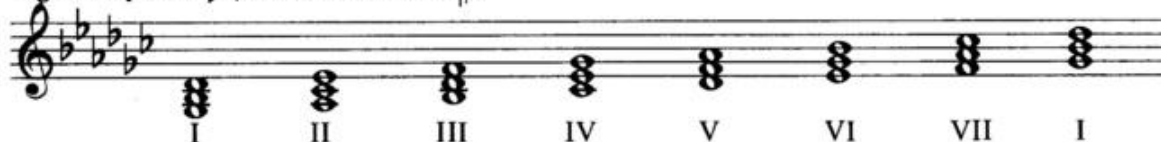


Fig. 9 Key of D $\flat$

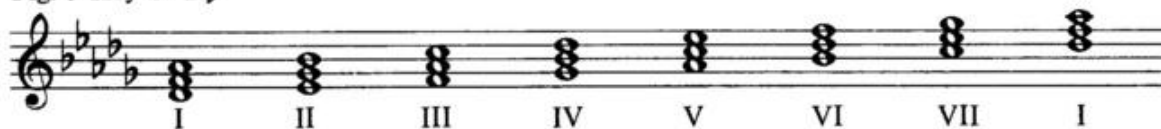


Fig. 10 Key of A $\flat$



Fig. 11 Key of E $\flat$



Fig. 12 Key of B $\flat$

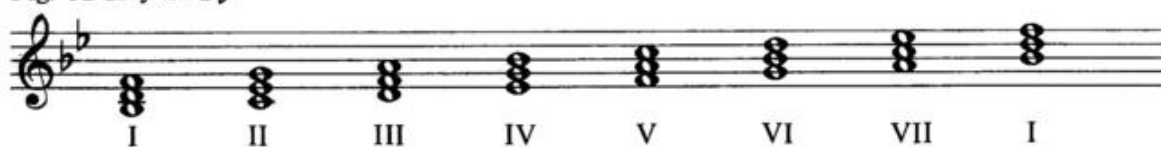


Fig. 13 Key of F



Intervals (Part 1)

These triads consist of three alternate scale tones sounded simultaneously.

The interval between the lowest and middle note is a 3rd; the interval between the lowest and highest note is a 5th.

Rule: If the upper note of a 3rd is contained within the diatonic major scale of the lower note, it is major (M). If it is lowered, it is minor (m). If it is raised, it is augmented (+).

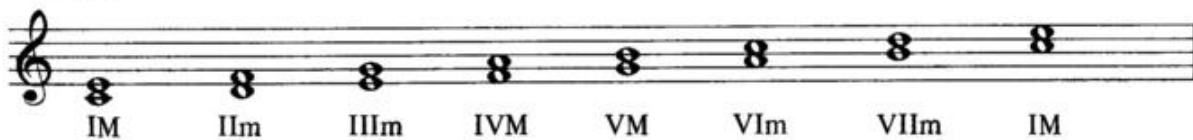
Fig. 14.



Thus, in the Key of C:

- I Based on the scale of C, C-E is major (M)
- II Based on the scale of D, D-F is minor (m)
- III Based on the scale of E, E-G is minor (m)
- IV Based on the scale of F, F-A is major (M)
- V Based on the scale of G, G-B is major (M)
- VI Based on the scale of A, A-C is minor (m)
- VII Based on the scale of B, B-D is minor (m)

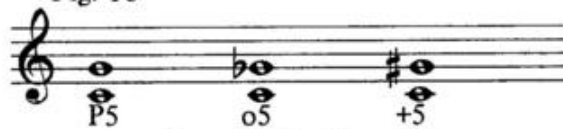
Fig. 15



This is true in all keys

Rule: If the upper note of a 5th is contained within the diatonic major scale of the lower note, it is perfect (P). If it is lowered, it is diminished (o). If it is raised, it is augmented (+).

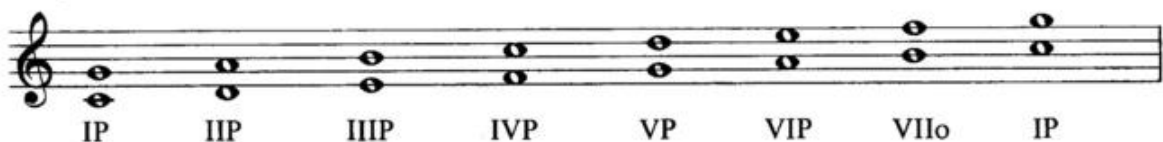
Fig. 16



Thus, in the Key of C:

- I Based on the scale of C, C-G is perfect (P)
- II Based on the scale of D, D-A is perfect (P)
- III Based on the scale of E, E-B is perfect (P)
- IV Based on the scale of F, F-C is perfect (P)
- V Based on the scale of G, G-D is perfect (P)
- VI Based on the scale of A, A-E is perfect (P)
- VII Based on the scale of B, B-F is diminished (o)

Fig. 17

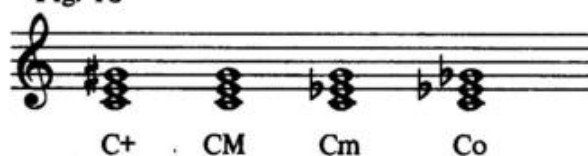


This is true in all keys

Combination		Position	Chord Quality
3rd	5th		
M	+	See note	augmented (+)
M	P	I - IV - V	major (M)
m	P	II - III - VI	minor (m)
m	o	VII	diminished (o)

Note: The augmented triad does not appear in the major diatonic system.(It does appear in the minor diatonic system.) It is included here because of its common usage in tunes.

Fig. 18



Rule: In any key, I, IV and V are major: II, III and VI are minor: VII is diminished. The augmented triad is added to the system.

Letters versus Numerals

Throughout this text the student will notice the interchangeable use of letters and numerals. Depending upon the specific problem under consideration, both are permissible. Illustrations, such as the Harmonic Vocabulary in which no key center is involved, obviously submit to lettered spelling; on the other hand, chord progressions or chord charts of tunes involving a key center demand the use of numerals in order to give the student a sense of key center unity which every experienced jazz musician uses as a natural process whether or not he or she is intellectually aware of the use of numerals. Numerals or figured bass (basso continuo) evolved during the Renaissance, and for some five centuries has been the classical vocabulary employed in music schools throughout the world. Ironically, improvisors in the classical tradition have historically employed numerals when performing in order to be free of the bondage of notation. Letters have never been employed (except for simple identification) in the classical field. The use of letters began in the early part of the of the present century; they were originally employed by the publishing industry to assist the average purchaser of sheet music in identifying ukulele or banjo chords. Since many of the early jazzmen were gifted but untrained musicians, they accepted the lettered chords as a readily available language in order to communicate with each other. However, when jazzmen perform, they "pre - hear" the next chord or phrase through a natural numerical estimate of where they are and where they are going. An added advantage involving numerals in progressions and chord charts should be apparent to the student. One set of numerals gives instantaneous twelve key facility since the numerals work for all keys. Otherwise, it would require twelve sets of letters to allow for transposition.

Chapter 2

The Seventh Chord System

Lesson: Play the scale - tone seventh chords of Fig. 1 through Fig. 13 in both hands until they can be played automatically from memory in any key.

Fig. 1 Key of C

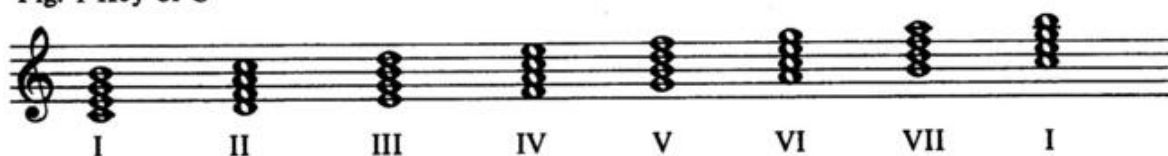


Fig. 2 Key of G

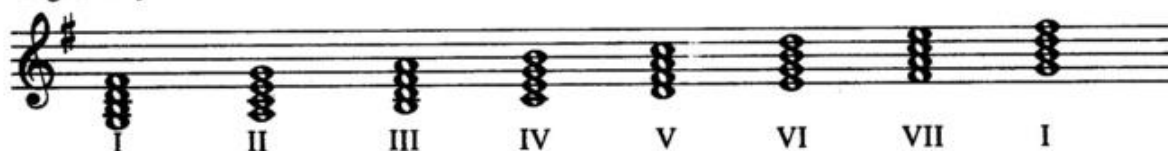


Fig. 3 Key of D

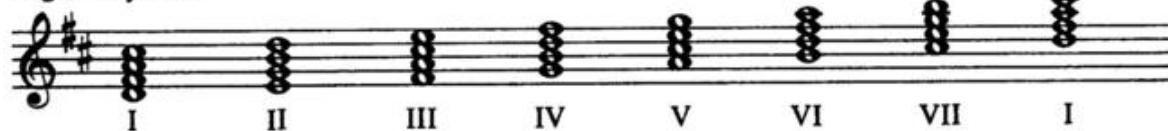


Fig. 4 Key of A

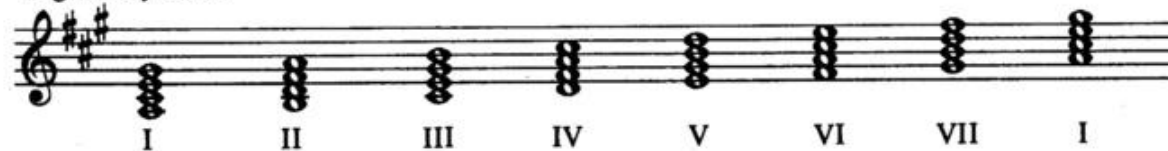


Fig. 5 Key of E

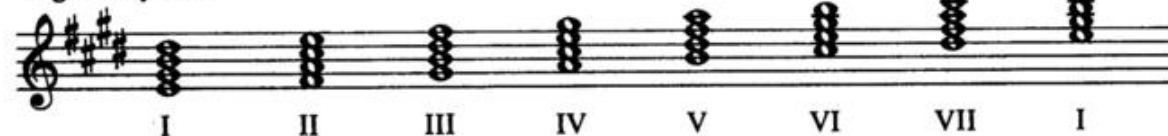


Fig. 6 Key of B

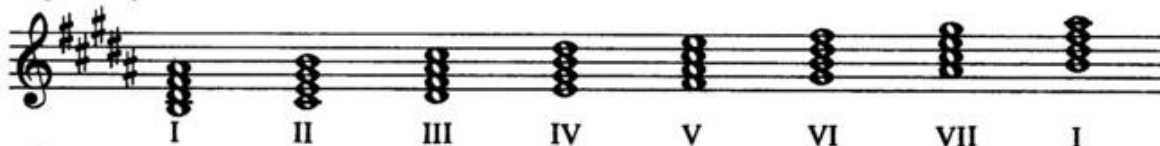


Fig. 7 Key of F# (Enharmonic of Gb)

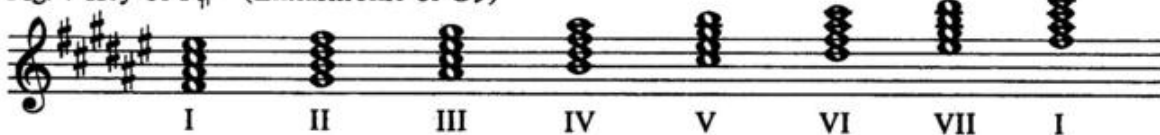


Fig. 8 Key of Gb (Enharmonic of F#)

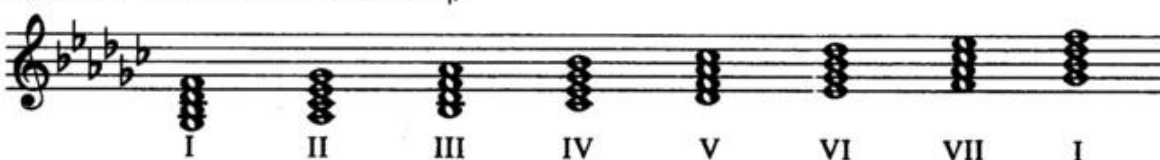


Fig. 9 Key of Db

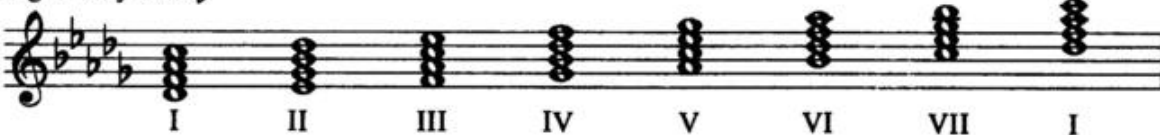


Fig. 10 Key of Ab



Fig. 11 Key of Eb

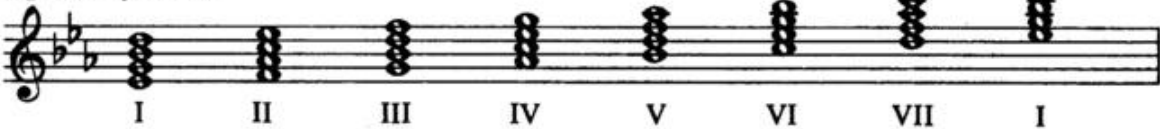


Fig. 12 Key of Bb

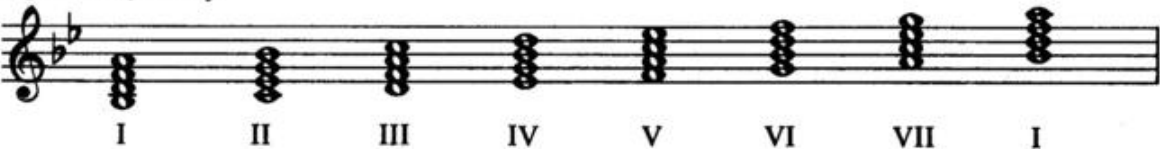
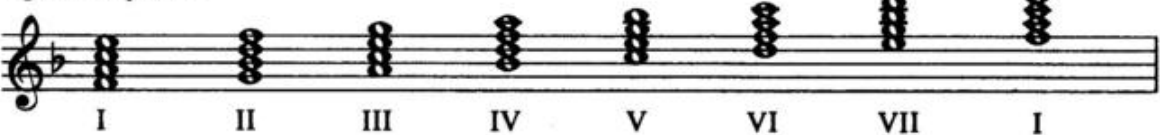


Fig. 13 Key of F



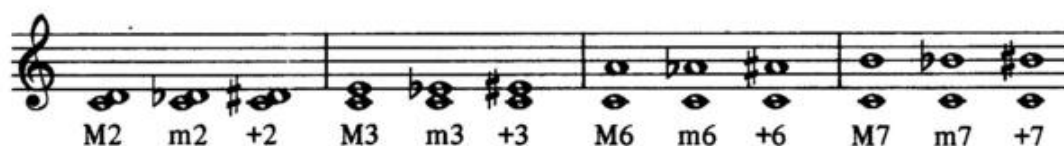
Chapter 3

Intervals (Part 2)

On Pages 7 and 8 rules for thirds and fifths were presented. The following indicates the general rule for all intervals:

If the upper note of a 2nd, 3rd, 6th or 7th is contained within the diatonic major scale of the lower note, it is major (M). If it is lowered, it is minor (m). If it is raised, it is augmented (+).

Fig. 1



If the upper note of a 4th, 5th or 8th (octave) is contained within the diatonic major scale of the lower note, it is perfect (P). If it is lowered, it is diminished (o). If it is raised, it is augmented (+).

Fig. 2



Fig. 3 Key of C



- I Based on the scale of C, C - E is major (M)
- II Based on the scale of D, D - F is minor (m)
- III Based on the scale of E, E - G is minor (m)
- IV Based on the scale of F, F - A is major (M)
- V Based on the scale of G, G - B is major (M)
- VI Based on the scale of A, A - C is minor (m)
- VII Based on the scale of B, B - D is minor (m)

- I Based on the scale of C, C - G is perfect (P)
- II Based on the scale of D, D - A is perfect (P)
- III Based on the scale of E, E - B is perfect (P)
- IV Based on the scale of F, F - C is perfect (P)
- V Based on the scale of G, G - D is perfect (P)
- VI Based on the scale of A, A - E is perfect (P)
- VII Based on the scale of B, B - F is diminished (o)

- I Based on the scale of C, C-B is major (M)
- II Based on the scale of D, D-C is minor (m)
- III Based on the scale of E, E-D is minor (m)
- IV Based on the scale of F, F-E is major (M)
- V Based on the scale of G, G-F is minor (m)
- VI Based on the scale of A, A-G is minor (m)
- VII Based on the scale of B, B-A is minor (m)

In all Keys:

Combination			Position	Chord Quality
3rd	5th	7th		
M	P	M	I - IV	major 7th
M	P	m	V	dominant 7th
m	P	m	II - III - VI	minor 7th
m	o	m	VII	half diminished 7th*

*Note: The term "half diminished" may be new to the student although it has wide usage in the classical nomenclature and, mainly through the efforts of the late Bill Evans, has begun to appear in contemporary jazz originals. The term is derived from the fact that the "full" diminished 7th chord (as we shall see) employs two diminished intervals (o5, o7), whereas the "half - diminished" chord employs only one diminished interval (o5).

Fig. 4



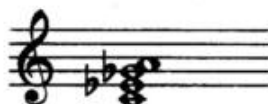
The Diminished Seventh Chord:

The diminished seventh chord does not appear in the major diatonic system. (It does appear in the minor diatonic system.) This chord contains a minor 3rd, a diminished 5th and a diminished 7th (lowered twice from the seventh scale position). (See Fig. 5)

Fig. 5



*Note: Although written as C - A, with the m 3rd and o 5th it remains a o7th.



The Sixty Chord System

Fig. 6. illustrates the Sixty Chord System used in jazz harmony. It consists of five different qualities which can be applied at any point on the keyboard.

There are twelve tones in the octave, each capable of supporting the five qualities; thus, the Sixty Chord System.

Transferring to sharps on m, ø, and o is for ease of "spelling" these chords.

Fig. 6

CM7 Cx7 Cm7 Cø7 Co7

DM7 Dx7 Dm7 Dø7 Do7

EM7 Ex7 Em7 Eø7 Eo7

F#M7 F#x7 F#m7 F#ø7 F#o7

GM7 Gx7 Gm7 Gø7 Go7

AM7 Ax7 Am7 Aø7 Ao7

Bm7 Bx7 Bm7 Bø7 Bo7

Enharmonic

DbM7 Dbx7 C#m7 C#ø7 C#o7

Ebm7 EbM7 Ebx7 Ebm7 D#ø7 D#o7

Fm7 Fx7 Fm7 Fø7 Fo7

Gbm7 Gbx7 F#m7 F#ø7 F#o7

AbM7 Abx7 Abm7 G#ø7 G#o7

Bbm7 Bbx7 Bbm7 A#ø7 A#o7

*Note: The use of the symbol x for the dominant chord is explained in the following chapter.

The Seventh Chord Symbols

The Major Seventh Symbol:

With the exception of the recent use of a triangle symbol (Δ), the traditional major symbol has always been either M7 or maj7. In this text the symbol M7 will be used.

The Dominant Seventh Symbol:

The traditional dominant symbols have always been 7 or 9 or 13 (C7, F9, etc.). The historical reason for this lay in the harmonic vocabulary of early jazz which employed the major triad, the minor triad, the diminished triad, the augmented triad and a fifth chord which demanded the use of the seventh, thus the 7 chord. Since all the seventh chords contain a seventh of some kind, it is apparent that the symbol 7 is improper. In this text the Juilliard symbol x7 will be used to indicate the dominant chord in any form (Cx7, Ebx7, Bbx9, etc.).

The Minor Seventh Symbol:

The traditional symbols for the minor seventh chord have been m7 or min.7. In this text the symbol m7 will be used.

The Half Diminished Symbol:

The half diminished symbol is only beginning to enter the public area. Traditionally it has been treated as minor 7♭5 (m7-5) or its first inversion minor 6 (m6). The symbol for half diminished is ϕ 7 and will be employed in this text.

The Diminished Seventh Symbol:

The traditional symbols for diminished have been a zero (o) or dim. The text will employ o7 in referring to both intervals and chords.

The Minor Sixth Symbol:

The term minor sixth (m6) can mean different things at various times, e.g. sometimes Cm6 means a minor triad with an added sixth. (See Fig. 1)

Fig. 1



But, in some notation, the term Cm6 can really mean Aø7. Cm6 (first inversion of Aø7) is the more familiar approximation of Aø7 or Am7^b5. In this text all root position ø7 chords will be indicated as such; the minor sixth chord will appear as minor added sixth (m⁺6) to avoid confusion with "6" as employed in inversions to be studied later. The same is true of major added sixth (M⁺6).

Symbol Summary:

M ^T	=	major triad
m ^T	=	minor triad
o ^T	=	diminished triad
+	=	augmented triad
M7	=	major seventh
x7	=	dominant seventh
m7	=	minor seventh
ø7	=	half diminished seventh
o7	=	diminished seventh
M ⁺ 6	=	major added 6th
m ⁺ 6	=	minor added 6th

Fig. 2 Key of C



*Note: The augmented triad is not indicated by "T" since the augmented chord does not appear in the seventh chord system.

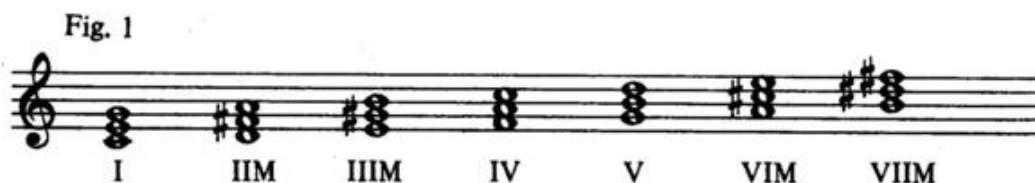
See the complete Harmonic Vocabulary in Chapter 8 for further illustration of the total system.

Altered Triads

The triads appearing in Chapter I and the seventh chords in Chapter 2 are shown in their primary functions; in other words, as the scale naturally forms them. If our use of harmony was limited to these primary functions we would be unable to perform even the simplest jazz material. Fortunately, the diatonic system readily lends itself to alteration. These alterations sub - divide into secondary functions and tertiary functions.

Secondary Functions:

Rule: A secondary function involves the raising or lowering of the 3rd and/or 5th of a triad; or the 3rd, 5th and/or 7th of a seventh chord to form a new chord or quality on any tone of the scale. (See Fig. 1)



In Fig. 1 the triads I - IV - V are naturally major; II - III - VI are minor which required raising the third in each case; VII is diminished which required raising both third and fifth in order to form major.

The following alteration table describes the intervals to be altered in order to form various triad qualities. (See Fig. 2)

Fig. 2

Quality M		Position I - IV - V	
Alteration	#5	b3	b3 b5
New Quality +		m	o
Quality m		Position II - III - VI	
Alteration	#3 #5	#3	b5
New Quality +		M	o
Quality o		Position VII	
Alteration	#3 x5	#3 #5	#5
New Quality +		M	m
Quality +		Position none	
Alteration	b5	b3 b5	b3 bb5
New Quality	M	m	o

n:

Complete the following triad qualities. Check your result against the Harmonic Vocabulary in Chapter 8.

C+ Gm Do AM Eb+ Bm F#m Gbo DbM Ab+ Ebm Bbo F+ Co

Tertiary Functions:

Rule:

A tertiary function involves raising or lowering an altered or unaltered chord chromatically out of the key. (See Fig. 4)

Key of C

I Io #Io
(secondary) (tertiary)

III IIIM bIIIM
(secondary) (tertiary)

V Vm bVm
(secondary) (tertiary)

VII VII+ bVII+
(secondary) (tertiary)

Rule:

In the chromatic harmonic system of jazz any chord can be built on any tone in the chromatic scale by altering, lowering or raising any chord in the diatonic system.

Chapter 6

Altered Seventh Chords

Triad harmony is used extensively in rock, country music and folk music as it was in Dixieland jazz from 1900 to 1920. In this period the only 7th chord employed was the dominant 7th but, gradually, through the twenties and into the early thirties, the seventh chord system began to replace the triad system until by 1934, with the development of the Goodman and Ellington bands, the seventh chord system with its extensive use of 9ths, 11ths and 13ths became the lasting language of jazz.

Secondary and Tertiary Functions:

As indicated in Chapter 5, the secondary and tertiary functions are formed by altering the 3rd, 5th and/or 7th or by lowering or raising the chord chromatically. See Fig. 1.

Fig. 1 Key of C

The figure displays three staves of music in the key of C, each showing a primary chord, its secondary function, and its tertiary function. The chords are represented by vertical stacks of notes on a five-line staff.

- Staff 1 (I):** Shows the I chord (C major), the I_o (secondary) chord (C minor), and the I_o (tertiary) chord (C major with a lowered 3rd, B $\flat$).
- Staff 2 (V):** Shows the V chord (G major), the V_m (secondary) chord (G minor), and the $\flat$ V_m (tertiary) chord (G minor with a lowered 3rd, E $\flat$).
- Staff 3 (VII):** Shows the VII chord (B major), the VII_x (secondary) chord (B major with a raised 3rd, D $\sharp$), and the $\flat$ VII_x (tertiary) chord (B major with a lowered 3rd, D $\flat$).

on:

Fig. 5 is an original tune employing the triad chromatic harmonic system.

The use of the symbol T will not be necessary since the piece consists entirely of triads.

Practice Fig. 5 until it can be played without hesitation.

Fig. 5

The musical score for Fig. 5 is presented in four systems. Each system consists of a treble staff and a bass staff. The treble staff contains a melody with eighth and quarter notes, and some measures have triplets indicated by a '3' over a bracket. The bass staff contains a harmonic line with triad symbols. The key signature has one flat (B-flat), and the time signature is 4/4. The piece is divided into four measures per system.

System 1:

- Measure 1: Treble staff has a triplet of eighth notes (G4, A4, B4) followed by a quarter rest. Bass staff has triad symbol II.
- Measure 2: Treble staff has a triplet of eighth notes (B4, C5, D5) followed by a quarter rest. Bass staff has triad symbol V.
- Measure 3: Treble staff has a quarter note (D5), an eighth note (C5), and a quarter rest. Bass staff has triad symbol III.
- Measure 4: Treble staff has a quarter note (B4), an eighth note (A4), and a quarter rest. Bass staff has triad symbol VI.

System 2:

- Measure 1: Treble staff has a quarter note (G4), an eighth note (F4), and a quarter rest. Bass staff has triad symbol IVm.
- Measure 2: Treble staff has a quarter note (E4), an eighth note (D4), and a quarter rest. Bass staff has triad symbol bIIIM.
- Measure 3: Treble staff has a quarter note (D4), an eighth note (C4), and a quarter rest. Bass staff has triad symbol bVI.
- Measure 4: Treble staff has a quarter note (B3), an eighth note (A3), and a quarter rest. Bass staff has triad symbol bIIIM.

System 3:

- Measure 1: Treble staff has a quarter note (G4), an eighth note (F4), and a quarter rest. Bass staff has triad symbol bVI.
- Measure 2: Treble staff has a quarter note (E4), an eighth note (D4), and a quarter rest. Bass staff has triad symbol bIIIM.
- Measure 3: Treble staff has a quarter note (D4), an eighth note (C4), and a quarter rest. Bass staff has triad symbol bVI.
- Measure 4: Treble staff has a quarter note (B3), an eighth note (A3), and a quarter rest. Bass staff has triad symbol bIIIM.

System 4:

- Measure 1: Treble staff has a quarter note (G4), an eighth note (F4), and a quarter rest. Bass staff has triad symbol bVI.
- Measure 2: Treble staff has a quarter note (E4), an eighth note (D4), and a quarter rest. Bass staff has triad symbol bIIIM.
- Measure 3: Treble staff has a quarter note (D4), an eighth note (C4), and a quarter rest. Bass staff has triad symbol bVI.
- Measure 4: Treble staff has a quarter note (B3), an eighth note (A3), and a quarter rest. Bass staff has triad symbol bIIIM.

First system of musical notation. The treble staff contains a melody in G major (one sharp). The bass staff contains Roman numerals indicating the harmonic progression: II, V, I, #Io, II, #IIo, III, III+, IV, #IVo.

Second system of musical notation. The treble staff continues the melody. The bass staff contains Roman numerals: v, bVo, IVm, III, bIIIo.

Third system of musical notation. The treble staff features a triplet of eighth notes in the third measure. The bass staff contains Roman numerals: II, III, IV, V, II, V.

Fourth system of musical notation. The treble staff continues the melody. The bass staff contains Roman numerals: III, VI, IVm, bIIIM, III, VIM, bVI, bIIIM.

Fifth system of musical notation. The treble staff continues the melody. The bass staff contains Roman numerals: bVIm, VIIM, III, VIM, II, V, I.

The following alteration table describes the intervals to be altered in order to form various seventh chord qualities.

Fig. 2

	Quality M7		Position I - IV	
Alteration	b7	b3 b7	b3 b5 b7	b3 b5 bb7
New Quality	x7	m7	ø7	o7

	Quality x7		Position V	
Alteration	#7	b3	b3 b5	b3 b5 b7
New Quality	M7	m7	ø7	o7

	Quality m7		Position II - III - VI	
Alteration	#3 #7	#3	b5	b5 b7
New Quality	M7	x7	ø7	o7

	Quality ø7		Position VII	
Alteration	#3 #5 #7	#3 #5	#5	b7
New Quality	M7	x7	m7	o7

	Quality o7		Position none	
Alteration	#3 #5 x7	#3 #5 #7	#5 #7	#7
New Quality	M7	x7	m7	ø7

Lesson:

Fig. 3 is an original tune employing the seventh chord chromatic harmonic system.

The use of the Arabic numeral 7 will not be necessary since there are no triads in the piece.

Practice Fig. 3 until it can be played without hesitation.

Fig. 3 Key of C

The musical score for Fig. 3 is presented in four systems, each with a treble and bass staff. The key signature is C major. The bass staff contains Roman numerals indicating the chords for each measure. The melody in the treble staff includes various rhythmic patterns, including triplets and slurs. A star symbol is placed under a note in the third measure of the first system.

System 1 (Measures 1-4):
 Treble: Measure 1 (quarter rest, eighth note, quarter note, eighth note), Measure 2 (quarter note, eighth note, quarter note, eighth note), Measure 3 (quarter note, eighth note, quarter note, eighth note), Measure 4 (quarter note, eighth note, quarter note, eighth note).
 Bass: Measure 1 (II $\emptyset$), Measure 2 (IV $\emptyset$), Measure 3 (III), Measure 4 (VI $\times$ $\flat$ 5).

System 2 (Measures 5-8):
 Treble: Measure 5 (quarter note, eighth note, quarter note, eighth note), Measure 6 (quarter note, eighth note, quarter note, eighth note), Measure 7 (quarter note, eighth note, quarter note, eighth note), Measure 8 (quarter note, eighth note, quarter note, eighth note).
 Bass: Measure 5 (bIII), Measure 6 (bVI $\times$ $\flat$ 5), Measure 7 (II), Measure 8 (III).

System 3 (Measures 9-12):
 Treble: Measure 9 (quarter note, eighth note, quarter note, eighth note), Measure 10 (quarter note, eighth note, quarter note, eighth note), Measure 11 (quarter note, eighth note, quarter note, eighth note), Measure 12 (quarter note, eighth note, quarter note, eighth note).
 Bass: Measure 9 (bVI), Measure 10 (bII $\times$ $\flat$ 5), Measure 11 (II $\emptyset$), Measure 12 (III).

System 4 (Measures 13-16):
 Treble: Measure 13 (quarter note, eighth note, quarter note, eighth note), Measure 14 (quarter note, eighth note, quarter note, eighth note), Measure 15 (quarter note, eighth note, quarter note, eighth note), Measure 16 (quarter note, eighth note, quarter note, eighth note).
 Bass: Measure 13 (IV $\emptyset$), Measure 14 (III), Measure 15 (VI $\times$ $\flat$ 5), Measure 16 (bIII).

* Note: The $\times$ $\flat$ 5 chords in measures 3, 4, 11, 12, 27 and 28 should be played with the anticipated (tied) figures; also the m chords in measures 5, 6, 13, 14, 29 and 30.

System 1: Treble staff contains a melodic line with various intervals and a triplet of eighth notes. Bass staff contains the following chords: II, III, IVm, Vm, bVI, and bIIx b5.

System 2: Treble staff contains a melodic line with a half note and a quarter note. Bass staff contains the following chords: I, I+6, Vm, IVm, bIII, and bII.

System 3: Treble staff contains a melodic line with a half note and a quarter note. Bass staff contains the following chords: II, bIIx, I, I+6, and Im.

System 4: Treble staff contains a melodic line with a half note and a quarter note. Bass staff contains the following chords: #II, #IVm, and VI.

System 5: Treble staff contains a melodic line with a half note and a quarter note. Bass staff contains the following chords: II, IV, III, VIx b5, bIII, and bVIx b5.

System 6: Treble staff contains a melodic line with a half note and a quarter note. Bass staff contains the following chords: II, III, IVm, Vm, bVI, bIIx b5, and I.

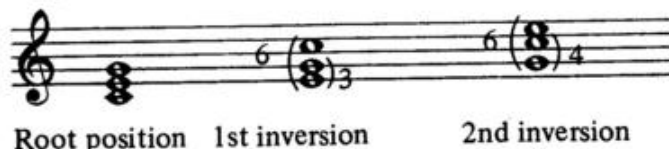
Chapter 7

Inversions

Inversions (Triads)

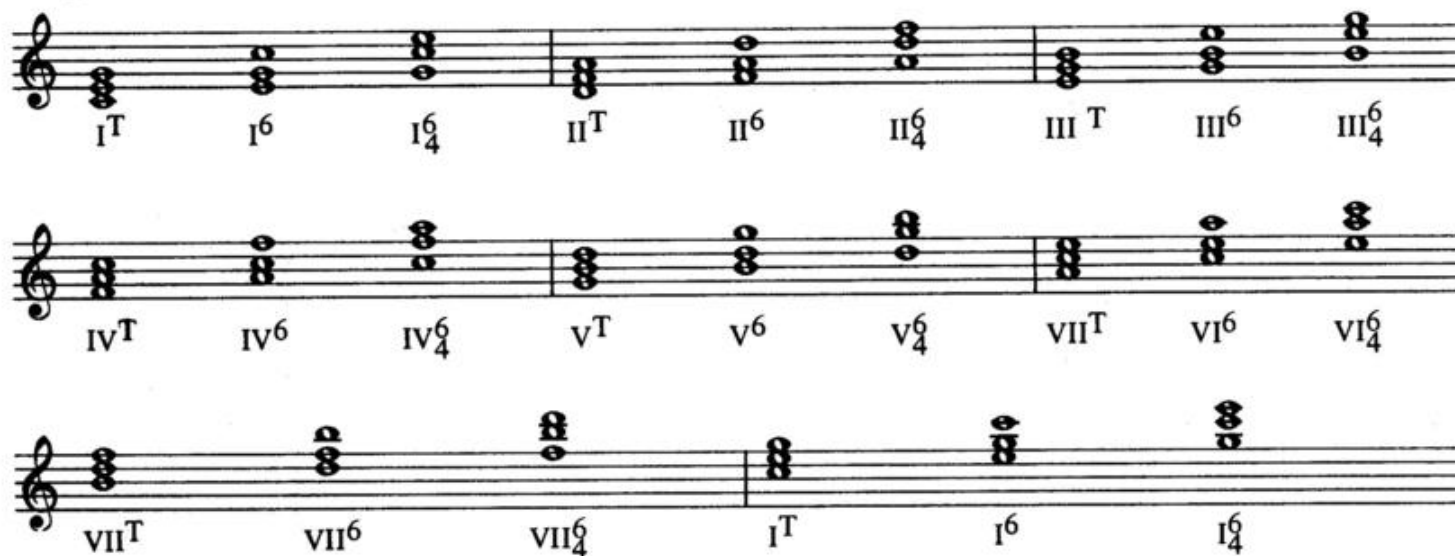
An inversion results from re-arranging the tones of a chord in such a way that the root of the chord is no longer in the bass. (See Fig. 1)

Fig. 1 Key of C



The traditional symbols for triad inversions are formed by counting the number of scale tones from the bass note to each of the remaining tones in the chord. Thus, in Fig. 1 the distance from E to C in the first inversion is a 6th; the distance from E to G is a 3rd. Thus, the symbol for first inversion is $\frac{6}{3}$ or, more usually, 6. In the second inversion the distance from G to E is a 6th; the distance from G to C is a 4th. Thus the symbol for second inversion is $\frac{6}{4}$ and that is never modified. Fig. 2 illustrates the triads of C with their inversions.

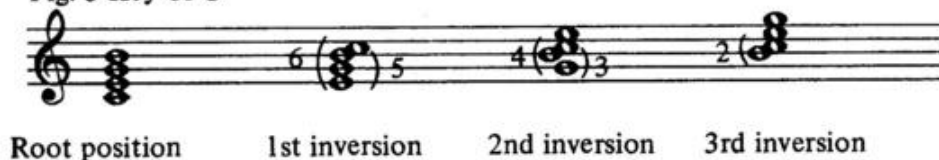
Fig. 2 Scale-tone triads of C and their inversions.



Lesson: Practice the scale-tone triads and their inversions in twelve keys - both hands.

Inversions (Seventh Chords)

Fig. 3 Key of C



Each inversion of a seventh chord contains an interval of a second (B - C in Fig.3). The symbol for the inversion is formed by counting the scale - tones from the bass note to each of the notes of the second. Thus, in Fig. 3, in the first inversion, E to C is a 6th; E to B is a 5th. Thus, the symbol is $\frac{6}{5}$. In the second inversion, G to C is a 4th; G to B is a 3rd. Thus, the symbol is $\frac{4}{3}$. In the third inversion B to C is a 2nd. Thus, the symbol is 2.

Fig. 4 illustrates the seventh chords of C with their inversions.

Fig. 4 Scale - tone seventh chords of C with their inversions.

The figure displays the scale-tone seventh chords of C and their inversions across four staves. Each staff contains eight chords, with a symbol below each indicating the intervals from the bass note.

- Staff 1:** I⁷, I $\frac{6}{5}$, I $\frac{4}{3}$, I₂, II⁷, II $\frac{6}{5}$, II $\frac{4}{3}$, II₂
- Staff 2:** III⁷, III $\frac{6}{5}$, III $\frac{4}{3}$, III₂, IV⁷, IV $\frac{6}{5}$, IV $\frac{4}{3}$, IV₂
- Staff 3:** V⁷, V $\frac{6}{5}$, V $\frac{4}{3}$, V₂, VI⁷, VI $\frac{6}{5}$, VI $\frac{4}{3}$, VI₂
- Staff 4:** VII⁷, VII $\frac{6}{5}$, VII $\frac{4}{3}$, VII₂, I⁷, I $\frac{6}{5}$, I $\frac{4}{3}$, I₂

Lesson: Practice the scale-tone seventh chords and their inversions in twelve keys - both hands.

Note: The student will note in Chapter 8 that the diminished seventh chord cannot be inverted since the tones are evenly spaced by minor thirds and no tone is adjacent to another. Of all the chords we have studied only the diminished seventh never loses its original intervals.

Chapter 8

The Harmonic Vocabulary

First system of musical notation showing the C major triad and its inversions. The treble clef staff contains three measures of chords: C major triad, C major first inversion, and C major second inversion. The bass clef staff contains the corresponding chord symbols: CM^T, CM⁶, and CM⁴.

Second system of musical notation showing the E major triad and its inversions. The treble clef staff contains three measures of chords: E major triad, E major first inversion, and E major second inversion. The bass clef staff contains the corresponding chord symbols: E⁺, E⁺, and E⁺.

Third system of musical notation showing the G major triad and its inversions. The treble clef staff contains three measures of chords: G major triad, G major first inversion, and G major second inversion. The bass clef staff contains the corresponding chord symbols: G⁺, G⁺, and G⁺.

Fourth system of musical notation showing the C minor triad and its inversions. The treble clef staff contains three measures of chords: C minor triad, C minor first inversion, and C minor second inversion. The bass clef staff contains the corresponding chord symbols: CM⁷, CM⁺, CM⁶, CM⁴, and CM₂.

Enharmonic

$D\flat M^T$ $D\flat m^6$ $D\flat m^4$ $D\flat m^T$ $D\flat m^6$ $D\flat m^4$ $C^\sharp m^T$ $C^\sharp m^6$ $C^\sharp m^4$

$C^\sharp o^T$ $C^\sharp o^6$ $C^\sharp o^4$ $C^\sharp +$ F^+ A^+

Enharmonic

$D\flat M^7$ $D\flat M^{+6}$ $C^\sharp M^{+6}$ $D\flat M^5$ $D\flat M^4$ $D\flat M^2$

Enharmonic

$D\flat x^7$ $D\flat x^6$ $D\flat x^4$ $D\flat x^2$ $C^\sharp x^7$ $C^\sharp x^6$ $C^\sharp x^4$ $C^\sharp x^2$

$C^\sharp m^7$ $C^\sharp m^{+6}$ $C^\sharp m^6$ $C^\sharp m^4$ $C^\sharp m^2$

System 1: Treble clef contains four chords with sharps. Bass clef contains the following chords: $C\sharp 7$, $C\sharp 6$, $C\sharp 4$, $C\sharp 2$, $C\sharp 7$, $E 7$, $G 7$, $B 7$.

System 2: Treble clef contains three chords with sharps. Bass clef contains the following chords: $DM 7$, $DM 6$, $DM 4$, $Dm 7$, $Dm 6$, $Dm 4$, $Do 7$, $Do 6$, $Do 4$.

System 3: Treble clef contains five chords with sharps. Bass clef contains the following chords: $D +$, $F \sharp$, $A \sharp$, $DM 7$, $DM + 6$, $DM 6$, $DM 4$, $DM 2$.

System 4: Treble clef contains five chords with sharps. Bass clef contains the following chords: $Dx 7$, $Dx 6$, $Dx 4$, $Dx 2$, $Dm 7$, $Dm + 6$, $Dm 6$, $Dm 4$, $Dm 2$.

System 5: Treble clef contains four chords with flats. Bass clef contains the following chords: $D 7$, $D 6$, $D 4$, $D 2$, $D 7$, $F 7$, $A\flat 7$, $B 7$.

Enharmonic

Staff 1: Treble clef. Chords: E-flat major triad, E-flat major 6th, E-flat major 4th. Staff 2: Bass clef. Chords: E-flat major triad, E-flat major 6th, E-flat major 4th, D-sharp minor triad, D-sharp minor 6th, D-sharp minor 4th.

Staff 1: Treble clef. Chords: D-sharp minor triad, D-sharp minor 6th, D-sharp minor 4th. Staff 2: Bass clef. Chords: E-flat augmented triad, G augmented triad, B augmented triad, E-flat major 7th, E-flat major 6th, E-flat major 5th, E-flat major 4th, E-flat major 2nd.

Staff 1: Treble clef. Chords: E-flat major 7th, E-flat major 6th, E-flat major 4th, E-flat major 2nd. Staff 2: Bass clef. Chords: E-flat major 7th, E-flat major 6th, E-flat major 5th, E-flat major 4th, E-flat major 2nd.

Enharmonic

Staff 1: Treble clef. Chords: D-sharp minor triad, D-sharp minor 6th, D-sharp minor 4th, D-sharp minor 2nd. Staff 2: Bass clef. Chords: D-sharp minor triad, E-flat major triad, F-sharp minor triad, A-flat major triad, C-flat major triad.

Staff 1: Treble clef. Chords: E major triad, E major 6th, E major 4th. Staff 2: Bass clef. Chords: E major triad, E major 6th, E major 4th, E-flat major triad, E-flat major 6th, E-flat major 4th.

System 1: Treble clef contains four measures of chords: E⁴G^{#4}, G^{#4}A⁴, A⁴B⁴, and B⁴C^{#5}. Bass clef contains five measures of chords: E⁺, G^{#+}, C⁺, EM⁷, EM⁺⁶, EM⁶₅, EM⁴₃, and EM².

System 2: Treble clef contains four measures of chords: E⁴G^{#4}, G^{#4}A⁴, A⁴B⁴, and B⁴C^{#5}. Bass clef contains five measures of chords: E^{x7}, E^{x6}₅, E^{x4}₃, E^{x2}, Em⁷, Em⁺⁶, Em⁶₅, Em⁴₃, and Em².

System 3: Treble clef contains four measures of chords: E⁴G^{#4}, G^{#4}A⁴, A⁴B⁴, and B⁴C^{#5}. Bass clef contains five measures of chords: E^{o7}, E^{o6}₅, E^{o4}₃, E^{o2}, E^{o7}, G^{o7}, B^{bb7}, D^{bb7}, and C^{#o7}. An "Enharmonic" bracket connects the last two measures of the bass clef.

System 4: Treble clef contains four measures of chords: E⁴G^{#4}, G^{#4}A⁴, A⁴B⁴, and B⁴C^{#5}. Bass clef contains five measures of chords: FM⁷, FM⁶, FM⁶₄, Fm⁷, Fm⁶, Fm⁶₄, Fo⁷, Fo⁶, and Fo⁶₄.

System 5: Treble clef contains four measures of chords: E⁴G^{#4}, G^{#4}A⁴, A⁴B⁴, and B⁴C^{#5}. Bass clef contains five measures of chords: F⁺, A⁺, C^{#+}, FM⁷, FM⁺⁶, FM⁶₅, FM⁴₃, and FM².

System 1: Treble clef staff shows chords in F major (F4, A4, C5) and F minor (F4, A4b, C5). Bass clef staff shows chord symbols: F_x^7 , F_x^6 , F_x^4 , F_x^2 , Fm^7 , Fm^{+6} , Fm^6 , Fm^4 , Fm^2 .

System 2: Treble clef staff shows chords in F major (F4, A4, C5) and F minor (F4, A4b, C5). Bass clef staff shows chord symbols: $F^{\flat 7}$, $F^{\flat 6}$, $F^{\flat 4}$, $F^{\flat 2}$, $F^{\flat o 7}$, $A^{\flat o 7}$, $B^{\flat o 7}$, $D^{\flat o 7}$.

System 3: Treble clef staff shows chords in F# major (F#4, A#4, C#5) and F# minor (F#4, A#4b, C#5). Bass clef staff shows chord symbols: $F^{\#}M^T$, $F^{\#}M^6$, $F^{\#}M^4$, $G^{\flat}M^T$, $G^{\flat}M^6$, $G^{\flat}M^4$, $F^{\#}m^T$, $F^{\#}m^6$, $F^{\#}m^4$. An "Enharmonic" bracket spans the first six chords.

System 4: Treble clef staff shows chords in F# major (F#4, A#4, C#5) and F# minor (F#4, A#4b, C#5). Bass clef staff shows chord symbols: $F^{\#}o^T$, $F^{\#}o^6$, $F^{\#}o^4$, $F^{\#}+$, $A^{\#}+$, D^+ .

System 5: Treble clef staff shows chords in F# major (F#4, A#4, C#5) and F# minor (F#4, A#4b, C#5). Bass clef staff shows chord symbols: $F^{\#}M^7$, $F^{\#}M^{+6}$, $F^{\#}M^6$, $F^{\#}M^4$, $F^{\#}M^2$, $G^{\flat}M^7$, $G^{\flat}M^{+6}$, $G^{\flat}M^6$, $G^{\flat}M^4$, $G^{\flat}M^2$. An "Enharmonic" bracket spans the first six chords.

Enharmonic

Chord symbols in bass clef: $F\sharp x^7$, $F\sharp x_5^6$, $F\sharp x_3^4$, $F\sharp x_2$, $G\flat x^7$, $G\flat x_5^6$, $G\flat x_3^4$, $G\flat x_2$, $F\sharp m^7$, $F\sharp m^{+6}$, $F\sharp m_5^6$, $F\sharp m_3^4$, $F\sharp m_2$

Chord symbols in bass clef: $F\sharp \emptyset^7$, $F\sharp \emptyset_5^6$, $F\sharp \emptyset_3^4$, $F\sharp \emptyset_2$, $F\sharp \emptyset^7$, $A\emptyset^7$, $C\emptyset^7$, $E\flat \emptyset^7$

Chord symbols in bass clef: Gm^T , Gm^6 , Gm_4^6 , Gm^T , Gm^6 , Gm_4^6 , Go^T , Go^6 , Go_4^6

Chord symbols in bass clef: $G+$, $B+$, $D\sharp+$, Gm^7 , Gm^{+6} , Gm_5^6 , Gm_3^4 , Gm_2

Chord symbols in bass clef: Gx^7 , Gx_5^6 , Gx_3^4 , Gx_2 , Gm^7 , Gm^{+6} , Gm_5^6 , Gm_3^4 , Gm_2

Treble clef: Four chords with flats (Bb, Ab, Gb, Fb).
 Bass clef: $G\flat^7$, $G\flat^6$, $G\flat^4$, $G\flat^2$, $G\flat^7$, $B\flat^7$, $D\flat^7$, $E\flat^7$

Treble clef: Four chords with flats, then three chords with sharps.
 Bass clef: $A\flat M^7$, $A\flat M^6$, $A\flat M^4$, $A\flat m^7$, $A\flat m^6$, $A\flat m^4$, $G\sharp m^7$, $G\sharp m^6$, $G\sharp m^4$
 Enharmonic

Treble clef: Four chords with sharps, then five chords with flats.
 Bass clef: $G\sharp o^7$, $G\sharp o^6$, $G\sharp o^4$, $G\sharp +$, $C +$, $E +$, $A\flat M^7$, $A\flat M^6$, $A\flat M^4$, $A\flat M^2$

Treble clef: Four chords with flats, then many chords with sharps.
 Bass clef: $A\flat x^7$, $A\flat x^6$, $A\flat x^4$, $A\flat x^2$, $A\flat m^7$, $A\flat m^6$, $A\flat m^4$, $A\flat m^2$, $G\sharp m^7$, $G\sharp m^6$, $G\sharp m^4$, $G\sharp m^2$

Treble clef: Four chords with sharps, then three chords with flats.
 Bass clef: $G\sharp o^7$, $G\sharp o^6$, $G\sharp o^4$, $G\sharp o^2$, $G\sharp o^7$, $B\flat^7$, $D\flat^7$, $F\flat^7$

AM^T AM⁶ AM⁶₄ Am^T Am⁶ Am⁶₄ Ao^T Ao⁶ Ao⁶₄

A+ C#+ F+ AM⁷ AM⁺⁶ AM⁶₅ AM⁴₃ AM² Ax⁷ Ax⁶₅ Ax⁴₃ Ax²

Am⁷ Am⁺⁶ Am⁶₅ Am⁴₃ Am² Ao⁷ Ao⁶₅ Ao⁴₃ Ao² Ao⁷ Co⁷ Ebo⁷ Gbo⁷ F#o⁷

Enharmonic

Bbm^T Bbm⁶ Bbm⁶₄ Bbm^T Bbm⁶ Bbm⁶₄ Bbo^T Bbo⁶ Bbo⁶₄ A#o^T A#o⁶ A#o⁶₄

Enharmonic

Bb+ D+ F#+ BbM⁷ BbM⁺⁶ BbM⁶₅ BbM⁴₃ BbM²

Enharmonic

B $\flat$ x⁷ B $\flat$ x⁶₅ B $\flat$ x⁴₃ B $\flat$ x² B $\flat$ m⁷ A $\sharp$ m⁷ B $\flat$ m⁺⁶ B $\flat$ m⁶₅ B $\flat$ m⁴₃ B $\flat$ m²

Enharmonic Enharmonic

A $\sharp$ $\emptyset$ ⁷ A $\sharp$ $\emptyset$ ⁶₅ A $\sharp$ $\emptyset$ ⁴₃ A $\sharp$ $\emptyset$ ² B $\flat$ $\emptyset$ ⁷ A $\sharp$ $\emptyset$ ⁷ D $\flat$ $\emptyset$ ⁷ C $\sharp$ $\emptyset$ ⁷ E $\emptyset$ ⁷ G $\emptyset$ ⁷

Bm^T Bm⁶ Bm⁶₄ Bm^T Bm⁶ Bm⁶₄ B $\emptyset$ ^T B $\emptyset$ ⁶ B $\emptyset$ ⁶₄

B⁺ D $\sharp$ ⁺ G⁺ Bm⁷ Bm⁺⁶ Bm⁶₅ Bm⁴₃ Bm² Bx⁷ Bx⁶₅ Bx⁴₃ Bx²

Bm⁷ Bm⁺⁶ Bm⁶₅ Bm⁴₃ Bm² B $\emptyset$ ⁷ B $\emptyset$ ⁶₅ B $\emptyset$ ⁴₃ B $\emptyset$ ² B $\emptyset$ ⁷ D $\emptyset$ ⁷ F $\emptyset$ ⁷ A $\flat$ $\emptyset$ ⁷

Note: From this point onward the use of the Arabic numeral 7 will no longer be necessary. All chords are seventh chords unless indicated otherwise.

Chapter 9

Jazz Rhythm

Improvisation in both Western and Eastern music has flourished for centuries; only since 1900 has improvisation in Western classical music declined, coinciding with the rise of jazz improvisation - a unique fusion of African rhythm and European harmony.

These unique qualities of jazz improvisation lie in the following:

1. The African device of accenting unstressed portions of the bar.
2. The specific rhythmic assignments usually employed as follows:

Rhythmic unit = Quarter note (♩)

Harmonic unit = Quarter note (♩)

Half note (♭)

Whole note (♩)

Melodic unit = Eighth notes (♪)

Eighth note triplets (♪)

Sixteenth notes (♩)

Sixteenth note triplets (♩)

The following historical treatments of the twelve - bar blues will illustrate how each style has employed these three units.

Chapter 10

Ragtime/Stride Piano 1900 - 1920

Rhythmic unit = ♩ (swing bass octaves)

Harmonic unit = ♩ and ♩ (combination of ♩ swing bass and ♩ "walk" bars)

Melodic unit = ♩ and ♩

The student is encouraged to seek out recordings of the following practitioners of this style:

"Jelly Roll" Morton

James P. Johnson

Willie "The Lion" Smith

"Fats" Waller (early)

Tom Turpin

James Scott

Scott Joplin

Fig. I. Key of B $\flat$

Moderately

Chorus 1

mf

(B $\flat$) Ix₂ VI_o II $\emptyset$ ₃⁴ I₄⁶ V₂

(B $\flat$) I I₂ Ix₂ IVx₅⁶ *P.N. IVx $\sharp$ IV_o

(B $\flat$) I₄⁶ IVx I₆⁶ $\flat$ III_o V₃⁴ V

(B $\flat$) V VI $\sharp$ VI_o V₅⁶ I Ix₂ VIx VIx₂ IIx₅⁶ V

* P.N. = passing note

Chorus 2

(Bb) I I₂ I_{x2} II⁴₃ II^{ø4}₃ I⁶₄ VI_o V⁶₅

(Bb) I I₂ I_{x2} I⁴₃ IV_x⁶₅ P.N. IV_x #IV_o

(Bb) I⁶₄ V₂ VI⁴₃ bIII_o II P.N. IV #IV_o

(Bb) V VI #VI_o V⁶₅ I I⁶ IV #IV_o V I

Swing Piano 1920 - 1940

Rhythmic unit =  (swing bass tenths)

Harmonic unit =  and  (combination of  swing bass and  "walk" bars)

Melodic unit = , , ,  and 

The student is encouraged to seek out recordings of the following stylists of this period:

"Fats" Waller (middle, late periods)

Earl Hines

Art Tatum

Teddy Wilson

Tenths

Tenths break down into three spans:

Span 1

C - E $\flat$
C $\sharp$ - E
F - A $\flat$
F $\sharp$ - A
G - B $\flat$
G $\sharp$ - B

Span 2

C - E
D - F
D $\sharp$ - F $\sharp$
E - G
F - A
G - B
A - C
B $\flat$ - D $\flat$
B - D

Span 3

D $\flat$ - F
D - F $\sharp$
E $\flat$ - G
E - G $\sharp$
F $\sharp$ - A $\sharp$
A $\flat$ - C
A - C $\sharp$
B $\flat$ - D
B - D $\sharp$

Spans 1 and 2 are within the normal hand span of the average pianist; span 3 is not and will be avoided in this text.

Fig. 1 illustrates swing piano: the first chorus is in the general style of Teddy Wilson, the second is more reminiscent of Art Tatum.

Fig. 1. Key of C

Chorus 1

The musical score for Chorus 1 is presented in four systems, each with a piano part and a corresponding harmonic analysis below it. The piano part is in 2/2 time and features a melodic line in the right hand and a harmonic line in the left hand. The harmonic analysis is written in Roman numerals and includes accidentals where necessary.

System 1:

(C) VI $\flat$ VI Ix_3^4 IVx $\sharp$ IVo Ix_3^4 $\flat$ Vo IVm

System 2:

(C) I^6 V_3^4 Ix IVx IVx_3^4 IVx $\sharp$ IVo

System 3:

(C) Ix_3^4 $\flat$ Vo $\flat$ VIIx $_3^4$ I^6 P.N. $\sharp$ Io II I^6 IV

System 4:

(C) V VI $\sharp$ VIo V_3^6 I $\flat$ VIIx $_3^4$ VIx_3^4 IIx_3^4 V

Chorus 2

(C) Ix Io II ϕ 2 I V/I

(C) Ix IVx V Ix IVx $\sharp$ IVo

(C) Ix $\frac{4}{3}$ $\flat$ VIIx $\frac{4}{3}$ VIx $\frac{4}{3}$ V $\frac{4}{3}$ $\sharp$ Io II VIx $\frac{4}{3}$ II $\frac{6}{3}$ $\sharp$ IVo

(C) V VI $\sharp$ VIo VII Ix Io II ϕ 2 I V I $\frac{4}{3}$ I

Chapter 12

Bop Piano 1940 - 1955

Rhythmic unit = ♩ (foot beat)

Harmonic unit = ♩ and ○ (left hand “shells”)

Melodic unit = ♩, ♩ and ♩

The important pianists of the early revolutionary period of bop were:

Earl “Bud” Powell
Thelonious Monk

Powell in particular forged a style which would accommodate the new demands of bop:

avoidance of swing bass
left hand shells (chord fragments)
right hand “horn line”

In the later period of bop the major influential pianists were:

Horace Silver
Hampton Hawes

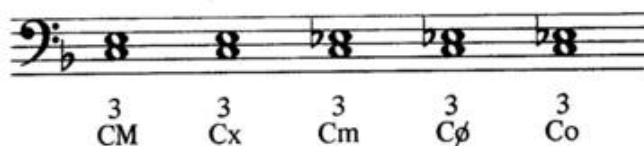
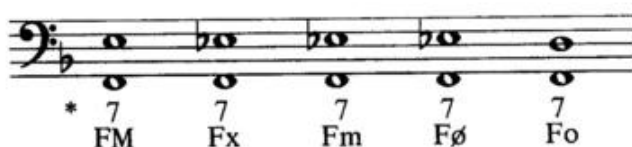
Shells

The use of shells are illustrated in Fig 1; they usually employ the root and the 7th or the root and the 3rd.

The fingerings usually employed are as follows:

Root and 7th = 5th finger and thumb.
Root and 3rd = 2nd finger and thumb.

Fig. 1 Key of F



*Note: The numerals above the chord names indicate the interval point of the left hand shell.

Since a shell can only imply the quality of a chord, the following table will be helpful in determining the function of each shell in Fig. 1.

Shell		Implied Quality
M 7th	=	M 7 chord
m 7th	=	$\left\{ \begin{array}{l} x 7 \text{ chord} \\ m 7 \text{ chord} \\ \emptyset 7 \text{ chord} \end{array} \right.$
o 7th	=	o 7 chord
M 3rd	=	$\left\{ \begin{array}{l} M 7 \text{ chord} \\ x 7 \text{ chord} \end{array} \right.$
m 3rd	=	$\left\{ \begin{array}{l} m 7 \text{ chord} \\ \emptyset 7 \text{ chord} \\ o 7 \text{ chord} \end{array} \right.$



Fig. 2 illustrates a drill which should be practiced around the circle of fifths from the twelve major thirds. The fingering is always 2 - 1 for 3rds and 5 - 1 for 7ths.

Fig. 2

Key of C

Left Hand Drill



Key of D $\flat$



Key of D



Key of E $\flat$



Key of E



Key of F



Key of G $\flat$



Key of G



Key of A $\flat$



Key of A



Key of B $\flat$



Key of B



Fig. 3 is a treatment of the twelve - bar blues in the characteristic style of bop piano.

Fig. 3 Key of B $\flat$

Lively

Introduction

mf

(B $\flat$) 7 III 7 $\flat$ III 7 II

Chorus 1

(B $\flat$) 7 II 7 $\flat$ IIx 7 I 3 IV 7 VII 3 IIIx 7 VI 3 IIx

(B $\flat$) 7 Vm 3 Ix 7 IVx 7 IVm 3 $\flat$ VIIx

(B $\flat$) 7 III 3 VIx 7 $\flat$ III 3 $\flat$ VIx 7 II

(Bb) 7 II 7 V 7 III 3 VIx 7 II 3 V

Chorus 2

(Bb) 7 I 3 IV 7 VII 3 IIIx 7 VI 3 IIx 7 Vm 3 Ix

(Bb) 7 IVx #7 IVo *6 VI₂ 7 IVx 7 III 7 bIII

(Bb) 7 II b7 IIx 7 I b7 VIIM

(Bb) 7 bVIM 7 Vb5 Ix Ix

* All inversion shells employ the outside voices of the particular inversion forming a point of six.

Chapter 13

Early Contemporary Jazz Piano 1955 - 1965

Rhythmic unit = ♩ (foot beat)

Harmonic unit = ♩ and ♩ (left hand voicings)

Melodic unit = ♩, ♩, ♩ and ♩

In the middle fifties stylistic changes in jazz piano began to be heard, particularly in the Miles Davis rhythm sections of the period. The important pianists in this early movement were:

"Red" Garland
Wynton Kelly

The basic problem of early contemporary jazz piano was to escape from the severe three voice style of bop piano into a more harmonic approach. This was achieved through the use of left hand voicings which will be thoroughly studied further on in this book.

Fig. 1 illustrates this style applied to the twelve - bar blues.

Fig. 1 Key of B $\flat$

Medium

Chorus 1

(B $\flat$) Ix

bIIx

Ix

(B $\flat$) bIIx

Ix

IVx

#IVx

IVx

(B $\flat$) Ix

VIIx

bVIIx

VIx

IIx

(B $\flat$) bIIx

Ix

bIIIx

IIx

bIIx

Chorus 2

(Bb) Ix IVx Ix bIIx

(Bb) Ix IVx

(Bb) IVx Ix VIIx bVIIx VIx

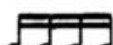
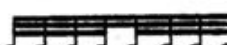
(Bb) IIx bIIx

(Bb) Ix bIIx Ix

Contemporary Jazz Piano 1965 - Present

Rhythmic unit =  (foot beat)

Harmonic unit =  and  (left hand voicings / modal fourth fragments)

Melodic unit = , ,  and 

Many of the further developments of contemporary jazz piano continued to occur in the Miles Davis rhythm sections with the appearance of Bill Evans, Herbie Hancock and Chick Corea. An exception was McCoy Tyner who achieved his innovations with John Coltrane.

The student should study carefully the recordings of:

Bill Evans
Herbie Hancock
Chick Corea
McCoy Tyner

Fig. 1 illustrates some of the basic devices employed by contemporary pianists as they would appear in the twelve - bar blues.

Fig. 1 Key of C

Brightly

Chorus 1

mf

(C) I $\sharp$ IIM $\sharp$ IV VIM IIIM I $\sharp$ IIM

(C) $\sharp$ IV IIIM Ix IVx IVx

(C) Ix VIIx $\flat$ VIIx VIx IIx

(C) $\flat$ IIx Ix $\flat$ IIIx IIx $\flat$ IIx

Chorus 2

(C) Ix $bIIIx$ bVx $bIIIx$ bVx $bIIIx$

(C) Ix IVx

(C) IVx Ix $bIIIx$

(C) IIx $bIIx$ Ix

(C) Ix Ix Ix

The student may notice the absence of several important pianists in the preceding summary, in particular Oscar Peterson and George Shearing. These omissions demand an explanation.

Oscar Peterson has established a level of virtuosity in jazz piano reminiscent of the great Art Tatum. Peterson's influences, however, are more modern - namely Bud Powell, Charlie Parker, George Shearing and Nat Cole. There are no major innovations in Peterson's playing as regards devices; his contribution lies in his brilliant consolidation of elements of the art from the time he first appeared on the jazz horizon in 1950.

George Shearing is a different matter. He is basically a Powell influenced pianist but is credited with developing and popularizing the block chord system previously associated with pianist Milt Buckner of the Lionel Hampton Band. This block chord system has become part of the jazz vernacular and is employed by many contemporary pianists in varying ways.

Two other pianists who left no heritage of devices employed by professional musicians but who brought jazz piano to the forefront of the American music public are Dave Brubeck and Erroll Garner. Both of these men were able to open new markets which had previously been largely inaccessible to jazz musicians. Brubeck, in colleges throughout the world; and Garner, through television and extensive recordings, brought a valuable exposure to jazz.

The Improvised Line

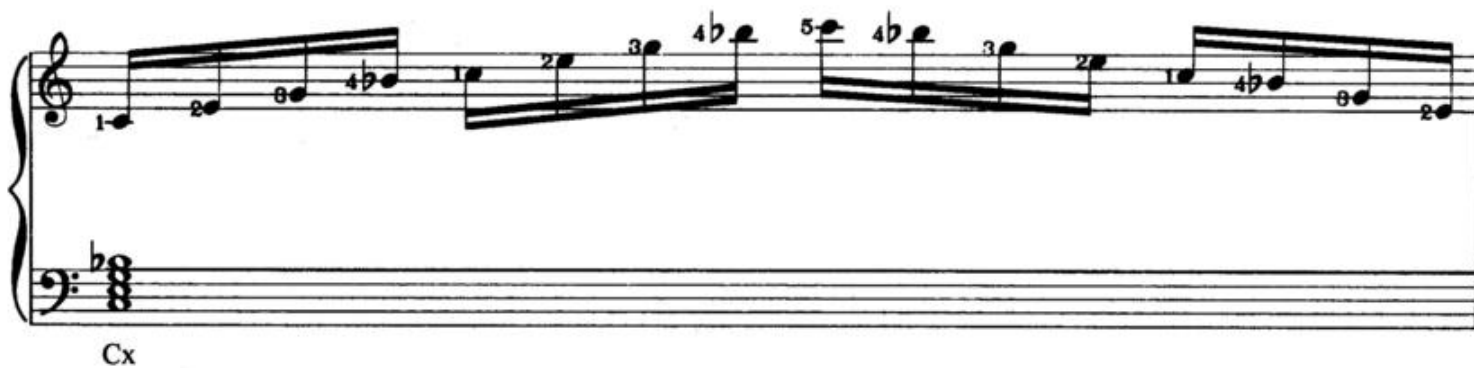
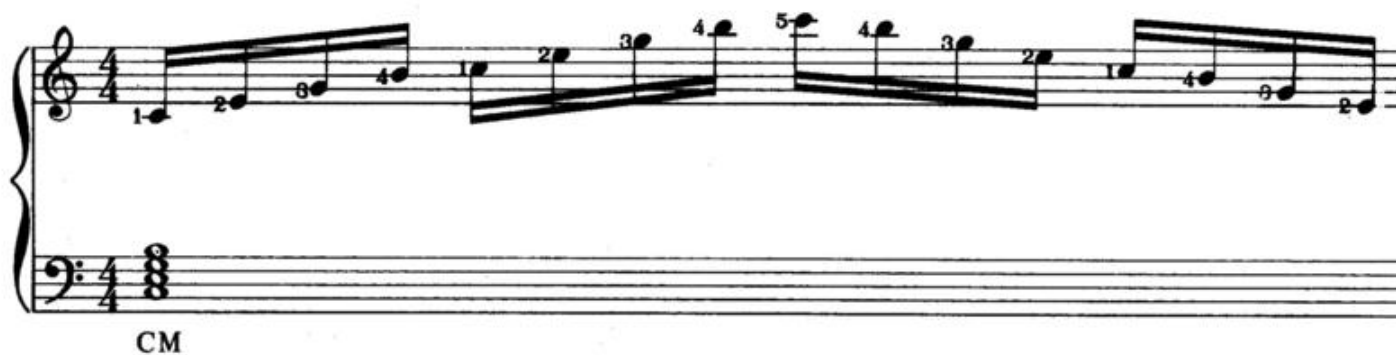
Jazz improvisation involves abandoning the melody and creating an improvised line based on the resources of the chords.

These resources consist of the following:

1. Arpeggios (broken chords)
2. Modes
3. Non - modal tones (blue notes)

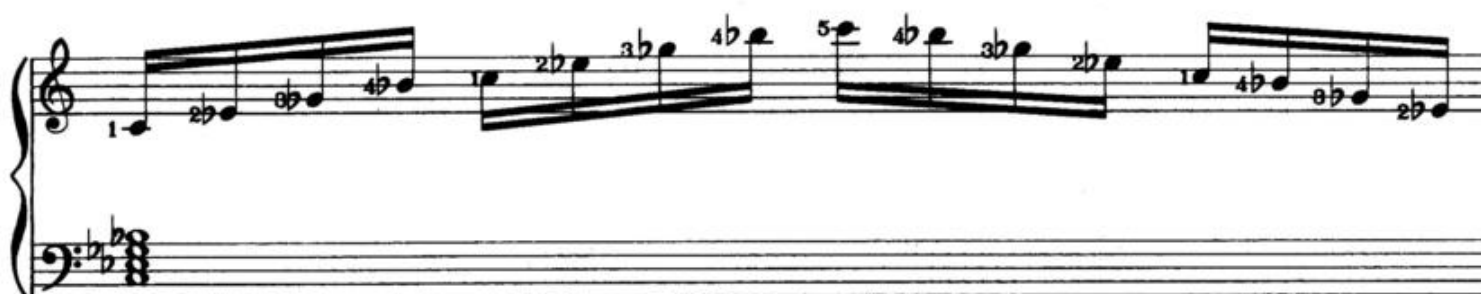
Arpeggios

Fig. 1 Key of C





Cm



Cø



Co

The following table illustrates the right hand arpeggio fingerings for the Sixty chords. (See Chapter Three)
Inversion fingerings are generally derived from root position fingerings.

Some fingerings employ only one fingering combination for all five qualities (M, x, m, ø, o); others require various fingerings as the intervals change.

The traditional rule concerning all fingering has been to avoid the thumb and, to some extent, the fifth finger on black notes in order not to disturb the classic piano hand position.

C	M, x, m, ϕ , o	123412345 - reverse
D	M, x, m, ϕ , o	123412345 - reverse
E	M, x, m, ϕ , o	123412345 - reverse
F	M, x, m, ϕ , o	123412345 - reverse
G	M, x, m, ϕ , o	123412345 - reverse
A	M, x, m, ϕ , o	123412345 - reverse
B	M, x, m, ϕ , o	123412345 - reverse
D $\flat$	M, x, m, ϕ , o	212341234 - reverse
A $\flat$	M, x, m, ϕ , o	212341234 - reverse
B $\flat$	M, x,	212341234 - reverse
(A $\sharp$)		
B $\flat$	m, ϕ , o	231234123 - reverse
(A $\sharp$)		
G $\flat$ (F $\sharp$)	M, x,	234123412 - reverse
G $\flat$ (F $\sharp$)	m, ϕ , o	212341234 - reverse
E $\flat$	M, x,	212341234 - reverse
E $\flat$	m,	123412345 - reverse
E $\flat$ (D $\sharp$)	ϕ , o	231234123 - reverse

Fig. 2 is an arpeggiated improvised line on "Here's That Rainy Day". Note the key changes.

The improvised line employs the ,  and  melodic units.

The left hand shells create a basic bop treatment in order to lend a sense of style to the study.

The student is advised to seriously study the arpeggio table appearing above for automatic facility.

Here's That Rainy Day

words and music by
Johnny Burke and James Van Heusen

Fig. 2 Key of G

The musical score is written for piano in 4/4 time, key of G major. It consists of four systems of music. The first system begins with a mezzo-piano (*mp*) dynamic marking. The melody is primarily in the right hand, with some triplets. The bass line provides harmonic support with chords. The second system continues the melody and bass line. The third system features a key change to E-flat major (indicated by two flats in the key signature). The fourth system continues in E-flat major. Chords are indicated by Roman numerals and figured bass notation below the bass staff.

Chord progressions (Bass staff):

- System 1: (G) 7 I, 7 VIIx#5, (Eb) III2, 7 IIx b5, 7 I
- System 2: (Eb) 7 IV, (G) 7 II, 7 III, 7 IV, 7 V b5, 7 I, 7 II
- System 3: (G) #7 IIo, 7 III, (Bb) x7 II, #7 II, 7 II, 3 V, 7 I, b7 Vx
- System 4: (Bb) 7 IV, 6 IV+6, (G) x7 II, #7 II, 7 II, 3 V, b7 VIIx, 7 VIx

(G) $\overset{7}{b}VIx$ $\overset{7}{V}$ $\overset{7}{I}$ $\overset{7}{VIIx}\sharp^5$ (Eb) $\overset{6}{III}_2$ $\overset{7}{b}IIx\flat^5$ (Eb) $\overset{7}{I}$

(Eb) $\overset{7}{IV}$ (G) $\overset{7}{II}$ $\overset{7}{III}$ $\overset{7}{IV}$ $\overset{7}{V}\flat^5$ $\overset{7}{I}$ $\overset{7}{VI}$

(G) $\overset{7}{Vm}$ $\overset{7}{b}Vx$ $\overset{7}{IV}$ $\overset{6}{VII}\frac{4}{3}$ $\overset{7}{IV}_o$

(G) $\overset{7}{III}$ $\overset{7}{b}III_o$ $\overset{7}{II}$ $\overset{7}{III}$

(G) $\overset{7}{IV}$ $\overset{7}{V}\flat^5$ $\overset{7}{I}\flat^5$ $\overset{6}{I}+6$

Chapter 16

Modes

A mode is a displaced scale starting on any tone in the row.

Thus, the scale of C can be played from:

- I C to C - Ionian Mode
- II D to D - Dorian Mode
- III E to E - Phrygian Mode
- IV F to F - Lydian Mode
- V G to G - Mixolydian Mode
- VI A to A - Aeolian Mode
- VII B to B - Locrian Mode

The terms appearing in the above table are the traditional Greek names applied to the modal displacements.

The basic problem here is to connect the eighty - four modes, (twelve scales, seven displacements) to the Sixty chord system in order to employ the most important device in contemporary jazz improvisation.

The Eighty-four Modes

(All fingerings follow the original Ionian fingering system)

Key of C

Ionian of C

Dorian of C

I

II

Detailed description: This block contains the first two modes. The Ionian of C (I) is shown in 4/4 time with a treble clef and a key signature of one sharp (F#). The melody starts on C4 and ascends stepwise to C5, with fingerings 1-2-3-4-5-4-3-2-1-3-2-1-3. The bass line consists of a single chord of C major. The Dorian of C (II) is shown in 4/4 time with a treble clef and a key signature of one flat (Bb). The melody starts on C4 and ascends stepwise to C5, with fingerings 2-1-3-4-5-4-3-2-1-3-2-1-3. The bass line consists of a single chord of C major.

Phrygian of C

Lydian of C

III

IV

Detailed description: This block contains the next two modes. The Phrygian of C (III) is shown in 4/4 time with a treble clef and a key signature of one flat (Bb). The melody starts on C4 and ascends stepwise to C5, with fingerings 9-1-2-3-4-5-4-3-2-1-3-2-1. The bass line consists of a single chord of C major. The Lydian of C (IV) is shown in 4/4 time with a treble clef and a key signature of two sharps (F# and C#). The melody starts on C4 and ascends stepwise to C5, with fingerings 1-2-3-4-5-4-3-2-1-3-2-1-3. The bass line consists of a single chord of C major.

Mixolydian of C

Aeolian of C

V

VI

Detailed description: This block contains the next two modes. The Mixolydian of C (V) is shown in 4/4 time with a treble clef and a key signature of one flat (Bb). The melody starts on C4 and ascends stepwise to C5, with fingerings 2-1-3-4-5-4-3-2-1-3-2-1-3. The bass line consists of a single chord of C major. The Aeolian of C (VI) is shown in 4/4 time with a treble clef and a key signature of two flats (Bb and Eb). The melody starts on C4 and ascends stepwise to C5, with fingerings 3-2-1-3-4-5-4-3-2-1-3-2-1. The bass line consists of a single chord of C major.

Locrian of C

VII

Detailed description: This block contains the seventh mode. The Locrian of C (VII) is shown in 4/4 time with a treble clef and a key signature of two flats (Bb and Eb). The melody starts on C4 and ascends stepwise to C5, with fingerings 4-1-2-3-4-5-4-3-2-1-3-2-1. The bass line consists of a single chord of C major.

Key of Db

Ionian of Db

I

Detailed description: This block contains the eighth mode. The Ionian of Db (I) is shown in 4/4 time with a treble clef and a key signature of three flats (Bb, Eb, and Ab). The melody starts on Db4 and ascends stepwise to Db5, with fingerings 2-1-3-4-5-4-3-2-1-3-2-1-3. The bass line consists of a single chord of Db major.

Dorian of D $\flat$ Phrygian of D $\flat$

II III

Lydian of D $\flat$ Mixolydian of D $\flat$

IV V

Aeolian of D $\flat$ Locrian of D $\flat$

VI VII

Key of D

Ionian of D Dorian of D

I II

Phrygian of D Lydian of D

III IV

Mixolydian of D

Aeolian of D

V VI

Detailed description: This block contains the first two scales. The Mixolydian of D scale (V) is shown in treble and bass staves with a key signature of two sharps (F# and C#). The Aeolian of D scale (VI) is shown in treble and bass staves with a key signature of one sharp (F#). Both scales are written in 4/4 time and include fingering numbers (1, 2, 3, 4, 5, 6, 7, 8).

Locrian of D

VII

Detailed description: This block contains the Locrian of D scale (VII) in treble and bass staves with a key signature of two sharps (F# and C#). The scale is written in 4/4 time and includes fingering numbers.

Key of E $\flat$

Ionian of E $\flat$

I

Detailed description: This block contains the Ionian of E $\flat$ scale (I) in treble and bass staves with a key signature of three flats (B $\flat$, E $\flat$, and A $\flat$). The scale is written in 4/4 time and includes fingering numbers.

Dorian of E $\flat$

Phrygian of E $\flat$

II III

Detailed description: This block contains the Dorian of E $\flat$ scale (II) and the Phrygian of E $\flat$ scale (III) in treble and bass staves with a key signature of three flats (B $\flat$, E $\flat$, and A $\flat$). Both scales are written in 4/4 time and include fingering numbers.

Lydian of E $\flat$

Mixolydian of E $\flat$

IV V

Detailed description: This block contains the Lydian of E $\flat$ scale (IV) and the Mixolydian of E $\flat$ scale (V) in treble and bass staves with a key signature of three flats (B $\flat$, E $\flat$, and A $\flat$). Both scales are written in 4/4 time and include fingering numbers.

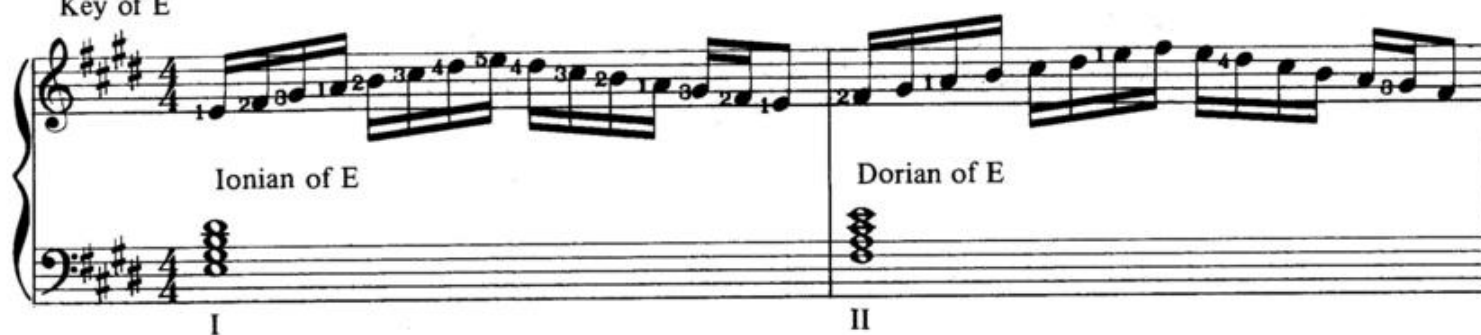
Aeolian of E $\flat$

Locrian of E $\flat$

VI VII

Detailed description: This block contains the Aeolian of E $\flat$ scale (VI) and the Locrian of E $\flat$ scale (VII) in treble and bass staves with a key signature of three flats (B $\flat$, E $\flat$, and A $\flat$). Both scales are written in 4/4 time and include fingering numbers.

Key of E



Ionian of E

Dorian of E

I II

Detailed description: This block contains the first two modes of the E major scale. The Ionian mode (I) is shown in the first measure, and the Dorian mode (II) is shown in the second measure. Both are in 4/4 time. The treble clef has a key signature of three sharps (F#, C#, G#). The bass clef shows the chord structure for each mode: Ionian is a triad of E, G#, and B; Dorian is a triad of E, G, and B.

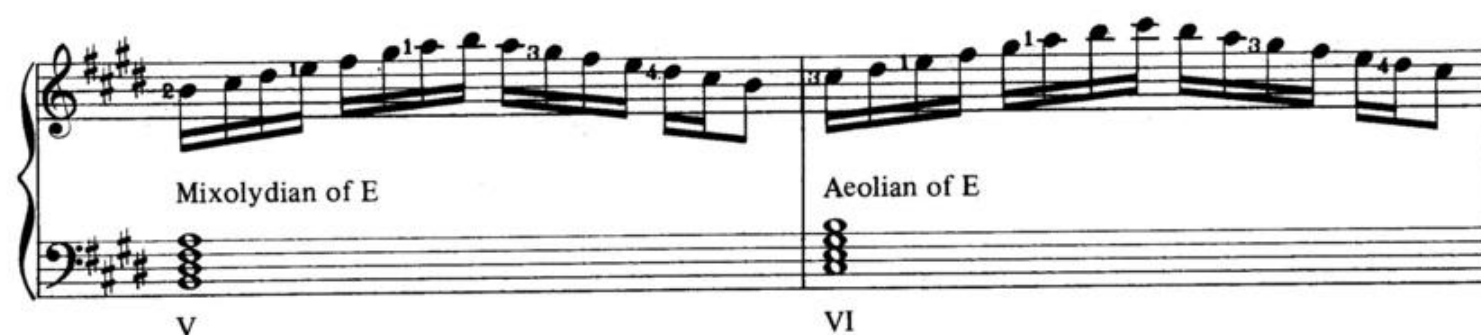


Phrygian of E

Lydian of E

III IV

Detailed description: This block contains the third and fourth modes of the E major scale. The Phrygian mode (III) is shown in the first measure, and the Lydian mode (IV) is shown in the second measure. Both are in 4/4 time. The treble clef has a key signature of three sharps (F#, C#, G#). The bass clef shows the chord structure for each mode: Phrygian is a triad of E, G, and Bb; Lydian is a triad of E, F#, and B.

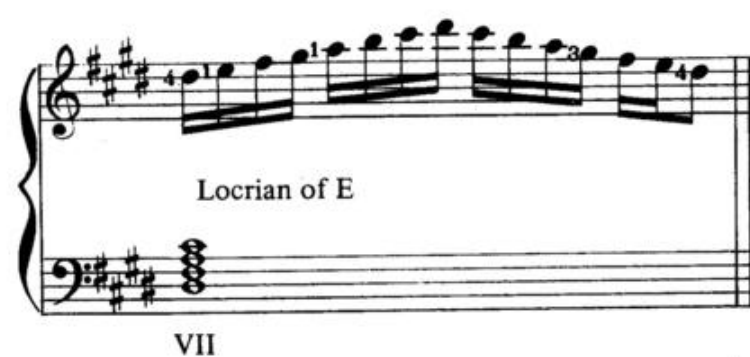


Mixolydian of E

Aeolian of E

V VI

Detailed description: This block contains the fifth and sixth modes of the E major scale. The Mixolydian mode (V) is shown in the first measure, and the Aeolian mode (VI) is shown in the second measure. Both are in 4/4 time. The treble clef has a key signature of three sharps (F#, C#, G#). The bass clef shows the chord structure for each mode: Mixolydian is a triad of E, G, and B; Aeolian is a triad of E, G, and Bb.

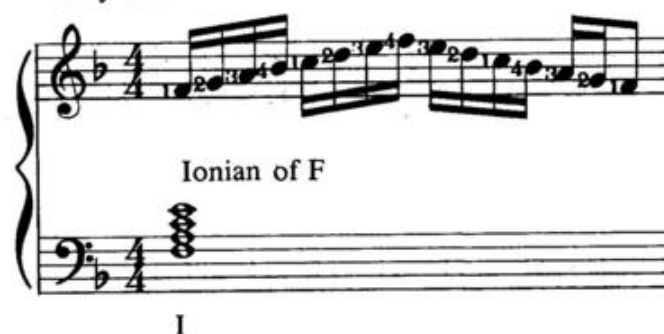


Locrian of E

VII

Detailed description: This block contains the seventh mode of the E major scale, the Locrian mode (VII). It is shown in the first measure. It is in 4/4 time. The treble clef has a key signature of three sharps (F#, C#, G#). The bass clef shows the chord structure for the mode: Locrian is a triad of E, G, and Bbb.

Key of F



Ionian of F

I

Detailed description: This block contains the first mode of the F major scale, the Ionian mode (I). It is shown in the first measure. It is in 4/4 time. The treble clef has a key signature of one flat (Bb). The bass clef shows the chord structure for the mode: Ionian is a triad of F, A, and C.



Dorian of F

Phrygian of F

II III

Detailed description: This block contains the second and third modes of the F major scale. The Dorian mode (II) is shown in the first measure, and the Phrygian mode (III) is shown in the second measure. Both are in 4/4 time. The treble clef has a key signature of one flat (Bb). The bass clef shows the chord structure for each mode: Dorian is a triad of F, A, and C; Phrygian is a triad of F, Ab, and C.

Lydian of F

Mixolydian of F

IV V

Aeolian of F

Locrian of F

VI VII

Key of F# (Enharmonic of Gb)

Ionian of F#

Dorian of F#

I II

Phrygian of F#

Lydian of F#

III IV

Mixolydian of F#

Aeolian of F#

V VI

Locrian of F $\sharp$

VII

Key of G $\flat$ (Enharmonic of F $\sharp$)

Ionian of G $\flat$

I

Dorian of G $\flat$

II

Phrygian of G $\flat$

III

Lydian of G $\flat$

IV

Mixolydian of G $\flat$

V

Aeolian of G $\flat$

VI

Locrian of G $\flat$

VII

Key of G

Ionian of G

I

Dorian of G

II



Phrygian of G

III

Lydian of G

IV

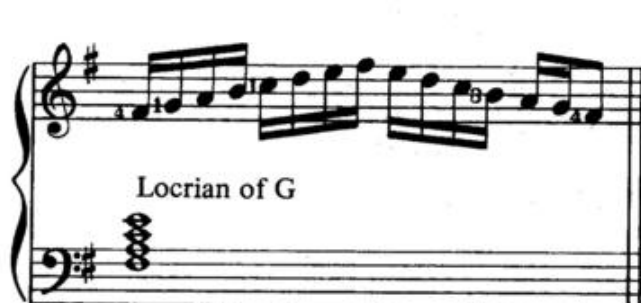


Mixolydian of G

V

Aeolian of G

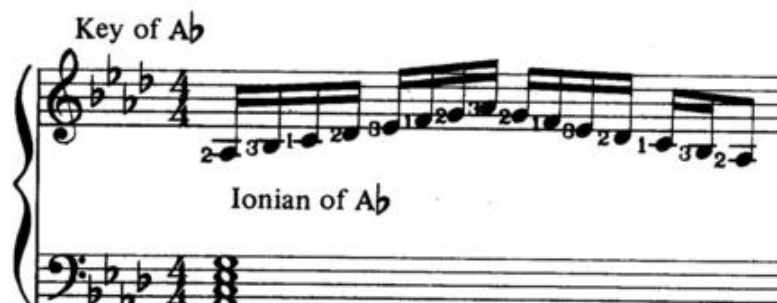
VI



Locrian of G

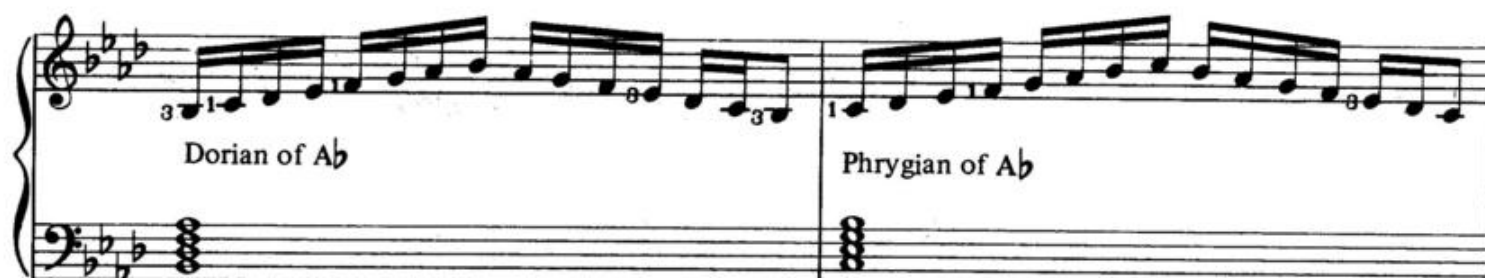
VII

Key of A $\flat$



Ionian of A $\flat$

I

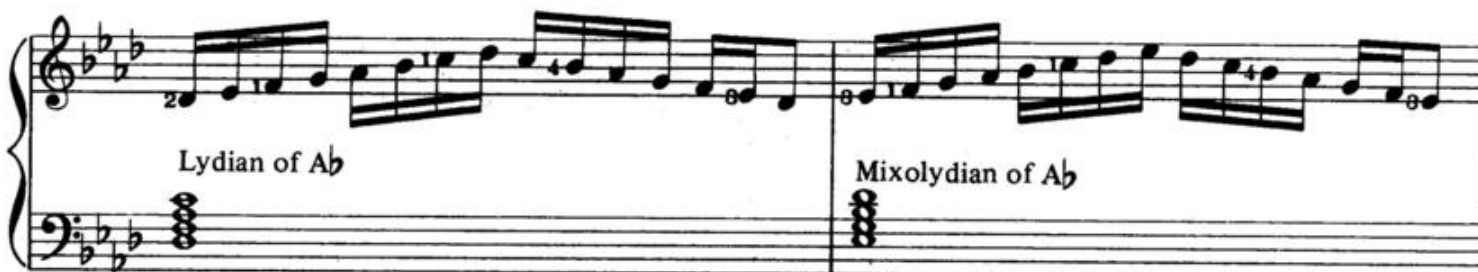


Dorian of A $\flat$

II

Phrygian of A $\flat$

III



Lydian of A $\flat$

IV

Mixolydian of A $\flat$

V

Aeolian of A $\flat$ Locrian of A $\flat$

VI VII

Detailed description: This block contains two musical systems. The first system, labeled 'VI', is for the Aeolian mode of A-flat. It features a treble clef with a key signature of three flats (B-flat, E-flat, A-flat) and a bass clef with a key signature of three flats. The melody in the treble clef consists of eighth notes: A-flat, B-flat, C, D, E-flat, F, G, A-flat. The bass clef contains a triad of A-flat, C, and E-flat. The second system, labeled 'VII', is for the Locrian mode of A-flat. It has the same key signature and bass triad. The melody in the treble clef consists of eighth notes: A-flat, B-flat, C, D, E-flat, F, G-flat, A-flat.

Key of A

Ionian of A Dorian of A

I II

Detailed description: This block contains two musical systems. The first system, labeled 'I', is for the Ionian mode of A. It features a treble clef with a key signature of three sharps (F-sharp, C-sharp, G-sharp) and a bass clef with a key signature of three sharps. The melody in the treble clef consists of eighth notes: A, B, C, D, E, F-sharp, G-sharp, A. The bass clef contains a triad of A, C-sharp, and E. The second system, labeled 'II', is for the Dorian mode of A. It has the same key signature and bass triad. The melody in the treble clef consists of eighth notes: A, B, C, D, E, F, G-sharp, A.

Phrygian of A Lydian of A

III IV

Detailed description: This block contains two musical systems. The first system, labeled 'III', is for the Phrygian mode of A. It features a treble clef with a key signature of three sharps and a bass clef with a key signature of three sharps. The melody in the treble clef consists of eighth notes: A, B-flat, C, D, E, F-sharp, G-sharp, A. The bass clef contains a triad of A, C-sharp, and E. The second system, labeled 'IV', is for the Lydian mode of A. It has the same key signature and bass triad. The melody in the treble clef consists of eighth notes: A, B, C, D, E, F-sharp, G, A.

Mixolydian of A Aeolian of A

V VI

Detailed description: This block contains two musical systems. The first system, labeled 'V', is for the Mixolydian mode of A. It features a treble clef with a key signature of three sharps and a bass clef with a key signature of three sharps. The melody in the treble clef consists of eighth notes: A, B, C, D, E, F-sharp, G, A. The bass clef contains a triad of A, C-sharp, and E. The second system, labeled 'VI', is for the Aeolian mode of A. It has the same key signature and bass triad. The melody in the treble clef consists of eighth notes: A, B, C, D, E, F, G, A.

Locrian of A

VII

Detailed description: This block contains one musical system labeled 'VII' for the Locrian mode of A. It features a treble clef with a key signature of three sharps and a bass clef with a key signature of three sharps. The melody in the treble clef consists of eighth notes: A, B-flat, C, D, E, F, G, A. The bass clef contains a triad of A, C-sharp, and E.

Key of B $\flat$

Ionian of B $\flat$

I

Detailed description: This block contains one musical system labeled 'I' for the Ionian mode of B-flat. It features a treble clef with a key signature of two flats (B-flat, E-flat) and a bass clef with a key signature of two flats. The melody in the treble clef consists of eighth notes: B-flat, C, D, E-flat, F, G, A, B-flat. The bass clef contains a triad of B-flat, D, and F.

Three measures of musical notation in Bb major. Each measure contains an ascending and descending scale in the treble clef and a triad in the bass clef. Measure 1: Dorian of Bb (Bb, C, D, Eb, F, G, A, Bb). Measure 2: Phrygian of Bb (Bb, Cb, D, Eb, F, G, A, Bb). Measure 3: Lydian of Bb (Bb, C, D, Eb, F#, G, A, Bb).

II

III

IV

Three measures of musical notation in Bb major. Each measure contains an ascending and descending scale in the treble clef and a triad in the bass clef. Measure 4: Mixolydian of Bb (Bb, C, D, Eb, F, G, Ab, Bb). Measure 5: Aeolian of Bb (Bb, Cb, Db, Eb, F, G, Ab, Bb). Measure 6: Locrian of Bb (Bb, Cb, Db, Eb, Fb, G, Ab, Bb).

V

VI

VII

Key of B

Two measures of musical notation in B major. Each measure contains an ascending and descending scale in the treble clef and a triad in the bass clef. Measure 7: Ionian of B (B, C#, D#, E, F#, G#, A#, B). Measure 8: Dorian of B (B, C, D, E, F#, G, A, B).

I

II

Two measures of musical notation in B major. Each measure contains an ascending and descending scale in the treble clef and a triad in the bass clef. Measure 9: Phrygian of B (B, C, D, E, F, G, A, B). Measure 10: Lydian of B (B, C#, D, E, F, G#, A, B).

III

IV

Three measures of musical notation in B major. Each measure contains an ascending and descending scale in the treble clef and a triad in the bass clef. Measure 11: Mixolydian of B (B, C#, D#, E, F#, G, A, B). Measure 12: Aeolian of B (B, C, D, E, F, G, A, B). Measure 13: Locrian of B (B, C, D, E, F, G, Ab, B).

V

VI

VII

Chapter 17

Modal Improvisation (Part 1)

Until 1940 the basic element employed in jazz improvisation was the arpeggio (broken chord). This was evolved by Louis Armstrong, Coleman Hawkins and a host of other gifted performers. Beginning in the late Thirties with the appearance of Lester Young, Charlie Parker, Fats Navarro and Bud Powell the use of displaced scales (modes) began and has continued to dominate the improvised line to the present day.

Now we have to connect the modal system described in Chapter Sixteen to the seventh chord harmonic system described in Chapter Two.

Fig. 1 illustrates the seventh chord system and the possible modal relationships.

Fig. 1

Quality	Intervals	Positions	Modes
Major 7	M P M	I	Ionian
		IV	Lydian
Dominant 7th	M P m	V	Mixolydian
Minor 7th	m P m	II	Dorian
		III	Phrygian
		VI	Aeolian
Half diminished 7th	m o m	VII	Locrian
Diminished 7th	m o o	none	none

The Major 7th

I - Ionian
IV - Lydian

The problem with the major chord lies in the fact that I is always I, but IV may be IV or I of a new key. Eventually, the decision as to which of the two modes to choose will be left to the student; but, for an initial drill all majors will be treated as I taking the Ionian mode.

Fig. 2 illustrates the twelve major chords with their accompanying Ionian modes. These modes should be practiced first one octave, then two, both ascending and descending for complete automatic facility.

The Major Scales

Fig. 2

Fig. 2 illustrates the twelve major chords with their accompanying Ionian modes. The figure shows four examples of major scales and their corresponding chords, arranged in two rows. Each example consists of a treble clef staff with a scale and a bass clef staff with a chord.

- Example 1: C Major (CM) scale and C Major (CM) chord.
- Example 2: D-flat Major (D $\flat$ M) scale and D-flat Major (D $\flat$ M) chord.
- Example 3: D Major (DM) scale and D Major (DM) chord.
- Example 4: E-flat Major (E $\flat$ M) scale and E-flat Major (E $\flat$ M) chord.

EM FM

(Enharmonic)

F#M GbM

GM AbM

AM BbM BM

The Dominant seventh

There is never any question about the status of the dominant chord since it only appears in the position of V. Fig. 3 illustrates the twelve dominant chords with their accompanying Mixolydian modes. These modes should be practiced first one octave, then two, both ascending and descending for complete automatic facility.

The Dominant Scales

Fig. 3

The figure displays twelve dominant scales and their corresponding dominant seventh chords, organized into four systems of three measures each. Each measure contains a Mixolydian scale in the treble clef and a dominant seventh chord in the bass clef. The scales are written in 4/4 time. The chords are labeled below each measure.

Scale	Chord
C Mixolydian	Cx(V of F)
D Mixolydian	Dx(V of G)
E Mixolydian	Ex(V of A)
F Mixolydian	Fx(V of B)
G Mixolydian	Gx(V of C)
A Mixolydian	Ax(V of D)
B Mixolydian	Bx(V of E)
C# Mixolydian	C#x(V of F#)
D# Mixolydian	D#x(V of G#)
E# Mixolydian	E#x(V of A#)
F# Mixolydian	F#x(V of B#)
G# Mixolydian	G#x(V of C#)
A# Mixolydian	A#x(V of D#)

The Minor 7th.

II - Dorian

III - Phrygian

VI - Aeolian

The minor chord is the most difficult to deal with since it appears in three positions (II - III - VI). For an initial drill the position of II (Dorian) will be used since the II position is the most commonly employed. Also the II chord forms an integral part of the basic cadence II - V - I (a basic anchor of all jazz improvisation). More advanced decisions should be left to the individual choice of the student.

Fig. 4 illustrates the twelve minor chords with their accompanying Dorian modes. These modes should be practiced first one octave, then two, both ascending and descending for complete automatic facility.

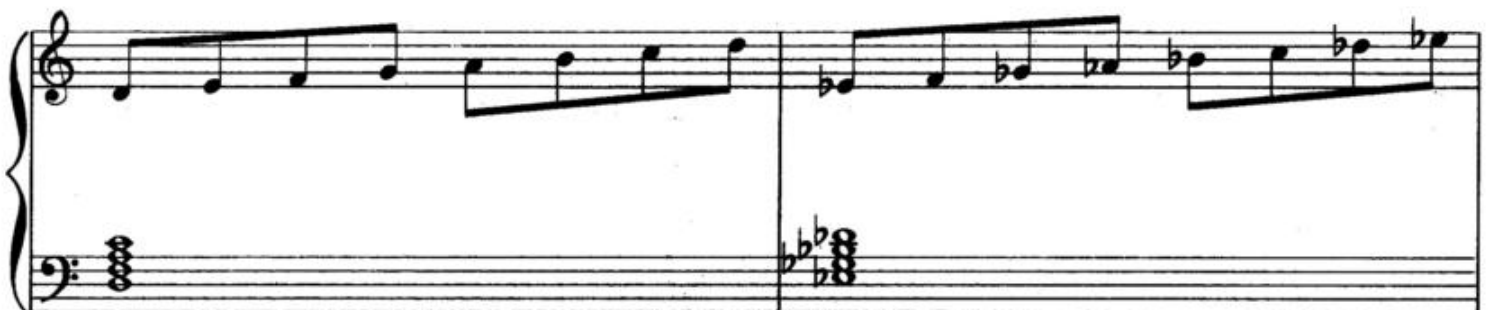
The Minor Scales

Fig. 4



Cm(II of B $\flat$)

C $\sharp$ m(II of B)



Dm(II of C)

E $\flat$ m(II of D $\flat$)

Em (II of D) Fm (II of E $\flat$)

F#m (II of E) Gm (II of F)

G#m (II of F#) A $\flat$ m (II of G $\flat$) Am (II of G)

B $\flat$ m (II of A $\flat$) Bm (II of A)

The Half Diminished Chord

Like the dominant chord, the half diminished chord appears in only one position (VII) and therefore always employs the Locrian mode.

Fig. 5 illustrates the twelve half diminished chords with their accompanying Locrian modes. These modes should be practiced first one octave, then two, both ascending and descending for complete automatic facility.

The Half Diminished Scales

Fig. 5

The figure displays twelve half diminished chords and their corresponding Locrian modes, organized into three systems of musical notation. Each system consists of a treble staff and a bass staff. The first system shows the Cø (VII of D♭) and C#ø (VII of D) chords and their modes. The second system shows the Dø (VII of E♭) and D#ø (VII of E) chords and their modes. The third system shows the Eø (VII of F), E#ø (VII of F#), and Fø (VII of G♭) chords and their modes, with an 'Enharmonic' bracket connecting the E#ø and Fø chords.

Cø (VII of D♭) C#ø (VII of D)

Dø (VII of E♭) D#ø (VII of E)

Eø (VII of F) E#ø (VII of F#) Fø (VII of G♭)

Enharmonic



F $\sharp$ ϕ (VII of G)

G ϕ (VII of A $\flat$)



G $\sharp$ ϕ (VII of A)

A ϕ (VII of B $\flat$)



A $\sharp$ ϕ (VII of B)

B ϕ (VII of C)

The Diminished Chord

The diminished chord presents a special problem since it does not appear naturally in the seventh chord system with which jazz is essentially concerned. To overcome this problem, jazz musicians have evolved two artificial scales which can be used interchangeably to accommodate the diminished chord. The two scales employ alternating major and minor seconds.

The Diminished Scales

Fig.6 and 7 illustrate the two alternating scales.

Fig. 6 Semitone combination: 021212121



2 indicates two half steps; 1 indicates one half step.

Fig. 7 Semitone combination: 012121212



Fig. 8 illustrates the twelve 021212121 diminished scales. The numbers indicate suggested fingerings.

Fig. 8



First system of musical notation showing two diminished scales. The left scale is labeled **Eo** and the right scale is labeled **Fo**. Both scales are written in treble and bass staves. The Eo scale starts on E4 and the Fo scale starts on F4. Both scales are 012121212 diminished scales. The numbers 1, 2, 3, 4, 5 indicate suggested fingerings for the right hand.

Second system of musical notation showing two diminished scales. The left scale is labeled **F#o** and the right scale is labeled **Go**. Both scales are written in treble and bass staves. The F#o scale starts on F#4 and the Go scale starts on G4. Both scales are 012121212 diminished scales. The numbers 1, 2, 3, 4, 5 indicate suggested fingerings for the right hand.

Third system of musical notation showing two diminished scales. The left scale is labeled **Ab o** and the right scale is labeled **Ao**. Both scales are written in treble and bass staves. The Ab o scale starts on Ab4 and the Ao scale starts on A4. Both scales are 012121212 diminished scales. The numbers 1, 2, 3, 4, 5 indicate suggested fingerings for the right hand.

Fourth system of musical notation showing two diminished scales. The left scale is labeled **Bb o** and the right scale is labeled **Bo**. Both scales are written in treble and bass staves. The Bb o scale starts on Bb4 and the Bo scale starts on B4. Both scales are 012121212 diminished scales. The numbers 1, 2, 3, 4, 5 indicate suggested fingerings for the right hand.

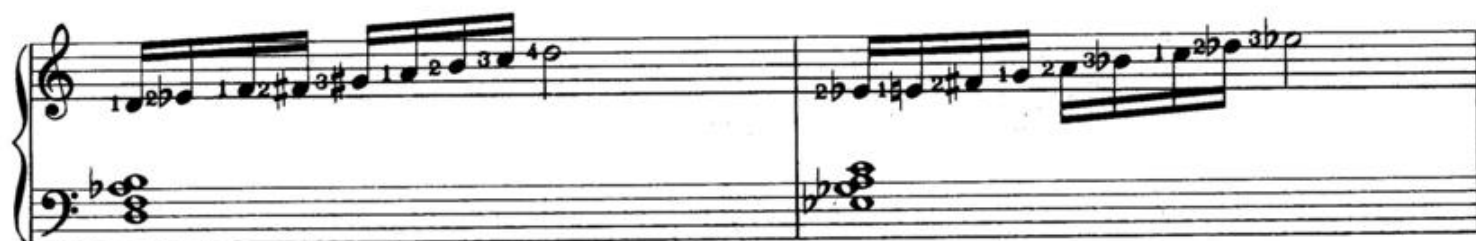
Fig. 9 illustrates the twelve 012121212 diminished scales. The numbers indicate suggested fingerings.

Fig. 9



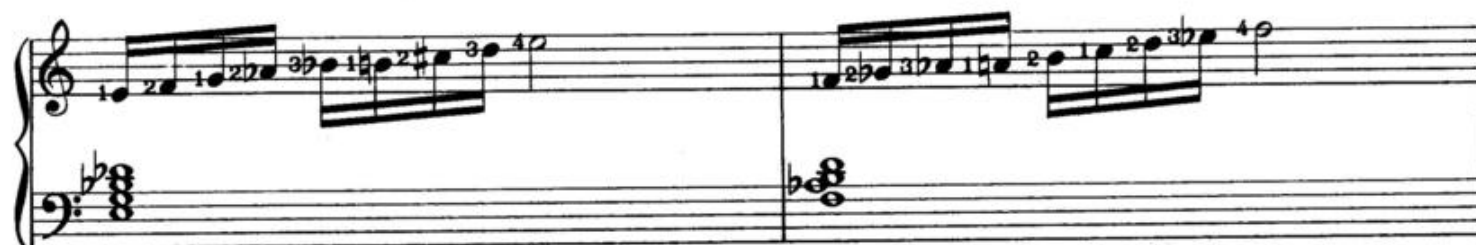
Co

C#o



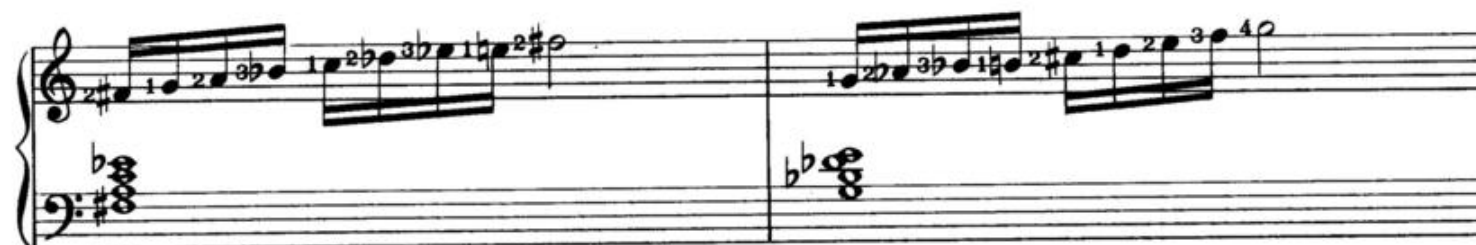
Do

Eb0



Eo

Fo



F#o

Go



G#o

Ao



B7o

Bo

Modal Improvisation (Part 2)

To complete the modal system we must deal with the three modes (Phrygian, Lydian, Aeolian) that were temporarily put aside in Chapter Seventeen.

The IV chord may be treated as IV or the I of a new key.
See Fig. 1

Fig. 1

Fig. 1 illustrates two different treatments of the IV chord. In the first measure, the IV chord (C major) is treated as the IV of C, and the mode is Lydian of C. In the second measure, the IV chord (F major) is treated as the I of a new key (F major), and the mode is Ionian of F.

What determines the status of the IV chord is the preceding harmony.
In Fig. 2, the harmony is centered around the I chord so that IV would assume its natural role and take the Lydian mode.

Fig. 2

Fig. 2 illustrates the natural role of the IV chord when the harmony is centered around the I chord. The sequence of chords is I (C major), II (D minor), III (E minor), and IV (F major). The mode is Lydian of C.

In Fig. 3, the key center around I has been weakened by the Vm and Ix and, as a result, the IV becomes the I of F.

As with all these modal conflicts, the student should make the decision concerning the "status" of the chord.

Fig. 3

Fig. 3 shows a musical sequence in 4/4 time. The bass line contains four chords: I (C major), Vm (G minor), Ix (D minor), and IV (F major). The IV chord is labeled "IV (I of F)". The treble line has a melodic line starting in the third measure, labeled "Ionian of F".

The III chord and the VI chord may be the III or VI of the original key or the II of a new key.

In Fig. 4, the strong tonal center establishes III as III and VI as VI.

Fig. 4

Fig. 4 shows a musical sequence in 4/4 time. The bass line contains six chords: I (C major), II (D minor), V (G major), III (E minor), and VI (F major). The III chord is labeled "Phrygian" and the VI chord is labeled "Aeolian". The treble line has a melodic line starting in the second measure.

In Fig. 5, the secondary functions in the case of III (IVm, $\flat$ VIIx) and VI (VIIIm, IIIx) establish the III as the II of D and the VI as the II of G.

Fig. 5

Fig. 5 shows two systems of musical notation, each with a treble and bass staff. The top system is labeled 'Dorian of D' and the bottom system is labeled 'Dorian of G'. Both systems show four measures of music. The chords indicated below the bass staff are:

- Top system: I, IVm, $\flat VIIx$, III (II of D)
- Bottom system: I, VIIm, IIIx, VI (II of G)

Fig. 6 illustrates the application of modal improvisation to "Polka Dots and Moonbeams" in the key of F. The III - IV and VI chords have been indicated as to status.

The bass line for Fig. 6 is as follows:

Introduction

(F) I VI / II $\flat Vm$ VIIx // I VI / IV II VIIm $\flat VIIx$ / VI II $\flat 4_3$ / VI₂ IV III $\flat$ III /
 (F) II / V VII₃ / III VI II V / I VI / II $\flat Vm$ VIIx /
 (F) I VI / IV II VIIm $\flat VIIx$ / VI II $\flat 4_3$ / VI₂ IV III $\flat$ III /
 (F) II $\flat$ IIx / I⁺⁶ (A) II ϕ $\flat$ IIx / I $\sharp$ Io / II V /
 (A) III VI / II $\flat$ IIx / I $\sharp$ Io / II V //
 (F) III VIx / II V / I VI / II $\flat Vm$ VIIx /
 (F) I VI / IV II VIIm $\flat VIIx$ / VI II $\flat 4_3$ / VI₂ IV III $\flat$ II / II $\flat$ IIx / I //

Polka Dots and Moonbeams

Fig. 6 Key of F

words and music by
Johnny Burke and James Van Heusen

Introduction
Slowly

Aeol.

I VI II bVIm VIIx

Aeol.

I VI IV II VIIIm bVIIx

Dor.

Aeol. Lyd. Phryg.

VI II^{b4}/₃ VI₂ IV III bIII II V VII⁴/₃

Phryg. Aeol.

Aeol.

III VI II V I VI II bVIm VIIx

Aeol. Lyd.

I VI IV II VII^m $\flat$ VII^x

Aeol. Lyd. Phryg.

VI II ϕ $\frac{4}{3}$ VI₂ IV III $\flat$ III

3

II $\flat$ II^x I⁺6 II ϕ $\flat$ II^x

3 3

Phryg. Aeol.

I $\sharp$ Io II V III VI

3 3

II $\flat$ II^x I $\sharp$ Io II V

Dor.

III VI^x II V

Aeol.

I VI II V^m VII^x

Aeol.

I VI Lyd. II VII^m VII^x

Dor.

VI II⁴ Aeol. Lyd. Phryg. VI₂ IV III bIII

II bII^x I I^{#6}

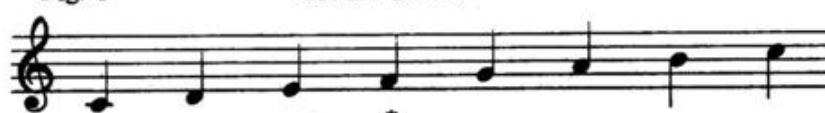
The Non-modal Tones

The non - modal tones, often referred to as “blue notes”, represent an important element in jazz improvisation since the tension created by these tones can add enormous excitement to a jazz performance.

It is often said that since Charlie Parker, there is no such thing as a “wrong” note, only the performer’s ability to make it sound “right”. However, the performer must keep a proper balance between the tension of the non - modal tones and the release of the modal tones.

Fig 1. illustrates the non - modal or, in the case of the diminished scale, non - scale tones relating to the five qualities.

Fig. 1

Modal Tones	Non - modal Tones
 C major	 C major
 C dominant	 C dominant
 C minor (Dorian)	 C minor
 C half diminished	 C half diminished
 C diminished	 C diminished

* The fourth step of both the major and dominant scales is often considered a tension tone unless the 3rd in the chord is raised to form an octave.

The non-modal tones in Fig. 1 break down into two categories

1. Passing tones
2. Ornamental tones

Passing tones	Ornamental tones
C major: C# D# A#	F# = augmented 11th G# (if also raised in chord to form M7#5)
C dominant: B	C# (Db) = flatted 9th D# = augmented 9th F# = augmented 11th G# (Ab) = flatted 13th
C minor: C# E F# G# B	
C half diminished: E G A B	D = 9th
C diminished: C# E G Bb	

The use of ornamental tones in harmony will be thoroughly studied in a succeeding chapter. Fig. 2 illustrates the use of both passing tones and ornamental tones as they apply to the improvised line on "Bye Bye Baby". The following is the bass line for "Bye Bye Baby" in Bb.

(Bb) I Io/I #Io / II II / II V /
 (Bb) I Io/I IV/ VII^{x7} VII^{#7} / VII^{x7} IIIx /
 (Bb) VI bVIo / Vm bV / IV / III VIx / II II^{x7} /
 (Bb) II bVIx / V IVx / III VIx / II V /
 (Bb) I Io / I #Io / II II^{x7} / II V /
 (Bb) I Io / I IV / VII^{x7} VII^{#7} / VI^{x7} IIIx /
 (Bb) VI bVIo / Vm bV / IV III / II #IIo /
 (Bb) III VIx / II V / I / I //

Bye, Bye Baby

Fig. 2

music by Jule Styne
words by Leo Robin

Brightly

The musical score consists of five systems of piano accompaniment. Each system contains a treble and bass staff. Chords are indicated by letters and accidentals below the bass staff. Some measures contain triplets, marked with a '3' and a bracket. The key signature is B-flat major (two flats).

System 1: (Bb) I, Io, I, #Io, x7 II, #7 II

System 2: (Bb) II, V, I, Io, I, IV

System 3: (Bb) x7 VIIm, #7 VIIm, VIIm, IIIx, VI, bVIIo

System 4: (Bb) Vm, bV, III, VIx, x7 II, #7 II

System 5: (Bb) II, bVIx, V, IVx, III, VIx

(B $\flat$) II V I Io I $\sharp$ Io

(B $\flat$) $\times$ 7 II $\sharp$ 7 II II V I Io

(B $\flat$) I IV $\times$ 7 VIIm $\sharp$ 7 VIIm VIIm IIIx

(B $\flat$) VI $\flat$ VIo Vm $\flat$ V IV III

(B $\flat$) II $\sharp$ IIo III VIx II V I I

Early Contemporary Left Hand Voicings

As was apparent in the transition from swing to bop piano (Chapters eleven and twelve) and again from bop to early contemporary (Chapters twelve and thirteen), revolutionary changes took place in the left hand structures employed in jazz piano.

Early contemporary left hand voicings began with two structures associated with classical piano. The earlier structure (referred to in this text as the (A) form) is usually attributed to Chopin and can be found extensively in his piano compositions. The more modern structure (referred to in this text as the (B) form) is usually attributed to the French impressionists, especially Maurice Ravel.

The following table describes the basic II-V-I procedure for the (A) form employing the Dorian, Mixolydian and Ionian modes.

II - 3572 - Dorian
V - 7236 - Mixolydian
I - 3562 - Ionian
See Fig. 1

The (A) Form

Fig. 1 Key of C

The figure displays a musical drill for the (A) form in the key of C, consisting of three measures. Each measure is divided into two staves: a grand staff (treble and bass clefs) and a single bass staff. The first measure is labeled '(C) II' and 'Dorian of C'. The second measure is labeled 'V' and 'Mixolydian of C'. The third measure is labeled 'I' and 'Ionian of C'. Above each grand staff, the chord voicings are shown with fingerings: for II, 2-7-6-3; for V, 6-3-2-7; and for I, 2-6-5-3. The bass staff shows a continuous eighth-note line starting from C2 and ascending to C4.

In the initial drill (Fig. 2), the voicings are played by the right hand in the middle C area; the root (which eventually would be assumed by the bass player in a group) is played by the left hand in the deeper range of the bass clef.

The II chord in each case creates a minor ninth chord; the V chord in each case creates a dominant nine thirteen chord; the I chord in each case creates a major ninth chord with an added sixth.

The (A) form will be studied in the keys of C, D $\flat$, D, E $\flat$, E and F.

This is to ensure that the register of the voicings is always around middle C.

Fig. 2 Drill
Key of C

(C) II V I

Key of D $\flat$

(D $\flat$) II V I

Key of D

(D) II V I

Key of E $\flat$

(E $\flat$) II V I

Key of E

(E) II V I

Key of F

(F) II V I

The following table describes the basic II - V - I procedure for the (B) form employing the Dorian, Mixolydian and Ionian modes.

II - 7 2 3 5 - Dorian

V - 3 1 3 7 2 - Mixolydian

I - 6 2 3 5 - Ionian

See Fig. 3

The (B) Form

Fig. 3 Key of G

The figure displays a musical exercise in the key of G, organized into three measures. The top staff is a grand staff (treble and bass clefs) with a 4/4 time signature. The middle staff is a single bass clef line. The bottom staff is a single bass clef line showing scale runs.

- Measure 1 (Dorian of G):** The grand staff shows a triad of G2, B2, and D3 in the bass clef. The middle staff has a whole note G2. The bottom staff shows an ascending scale from G2 to D3.
- Measure 2 (Mixolydian of G):** The grand staff shows a triad of G2, B2, and F#3 in the bass clef. The middle staff has a whole note G2. The bottom staff shows an ascending scale from G2 to D3.
- Measure 3 (Ionian of G):** The grand staff shows a triad of G2, B2, and D3 in the bass clef. The middle staff has a whole note G2. The bottom staff shows an ascending scale from G2 to D3.

Labels (G), II, V, and I are placed below the middle staff in measures 1, 2, and 3 respectively. Labels "Dorian of G", "Mixolydian of G", and "Ionian of G" are placed below the bottom staff in measures 1, 2, and 3 respectively.

In the initial drill (Fig. 4), the (B) form will be studied in the keys of G, A $\flat$, A, B $\flat$, B and F $\sharp$. This is to ensure that the register of the voicings is always around middle C.

Fig. 4 Drill

Key of G

(G) II V I

Key of A $\flat$

(A $\flat$) II V I

Key of A

(A) II V I

Key of B $\flat$

(B $\flat$) II V I

Key of B

(B) II V I

Key of F $\sharp$

(F $\sharp$) II V I

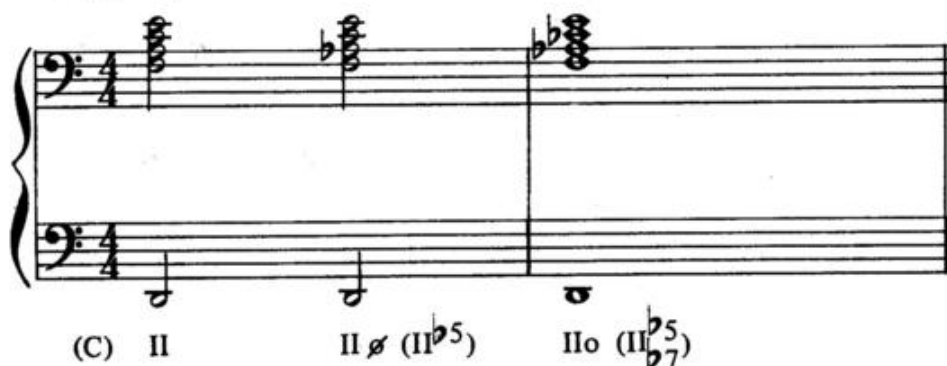
The use of II - V - I is due to its elemental role in jazz improvisation.
The problem now is to break open the II - V - I capsule in order to
 accommodate the Sixty Chord demands of jazz

Therefore:

All major chords are I
 All dominant chords are V
 All minor chords are II
 All half diminished chords are II $\flat 5$
 All diminished chords are II $\flat 5 \flat 7$ } See note

Note: In the case of half diminished and diminished chords, the voicings are derived from the chord (minor) with the most similar intervals to ϕ and o. (See Fig. 5)

Fig. 5 Key of C

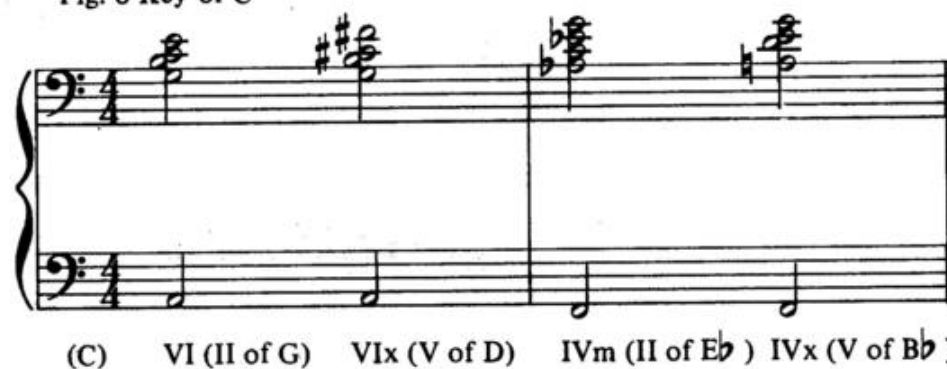


If the student attempts to play the (A) form in the (B) form keys or vice versa, it will be noticed immediately that the voicing is too low or too high and removed from the middle C area. To avoid this the following rule is applied:

If the parent key of the chord occurs between C and F (C, D $\flat$, D, E $\flat$, E, F), use the (A) form; if the parent key occurs between F $\sharp$ and B (F $\sharp$, G, A $\flat$, A, B $\flat$, B), use the (B) form.

For example, VI in the key of C (a minor chord) becomes the temporary II of G and takes the (B) form (G is a (B) form key). On the other hand, VIx in the key of C (a dominant chord) is the temporary V of D and takes the (A) form (D is an (A) form key). (See Fig. 6)

Fig. 6 Key of C



The use of voicings in this manner automatically offers the student the following:

- Proper register (keyboard area)
- Proper voice - leading
- Automatic ornamentation
- Contemporary sound

Author's note; It is absolutely essential that the student memorize an automatic facility with the twelve II - V - I patterns in Figs. 2 and 4.

The Basic Theory of Intervals



In Fig. 7 the basic theory of all intervals is illustrated.

Intervals are traditionally conceived in alternate steps with odd numbers (1, 3, 5, 7, etc.); however, when building structures, 9 can be identified for convenience as 2, 11 as 4 and 13 as 6; but these tones still retain their original status or function.

The exception to this is the sixth tone which, as the following table will indicate, takes on different values depending upon the quality of the particular chord.

The Sixth Tone:

- Major 7th chord = added 6th
- Dominant 7th chord = 13th
- Minor 7th chord = added 6th
- Half diminished 7th chord = non-functioning
- Diminished 7th chord = 7th of the chord

The unique quality of the 13th in the dominant chord results from the combination of the major 3rd and the minor 7th in the dominant chord.

Contemporary Left Hand Voicings Scales

Figures 1 through 13 illustrate the voicing system through the twelve major scales. Note that the II - V - I parent key system has been used in order to determine the (A) or (B) form status of each voicing.

Fig. 1 Key of C



Fig. 2 Key of Db

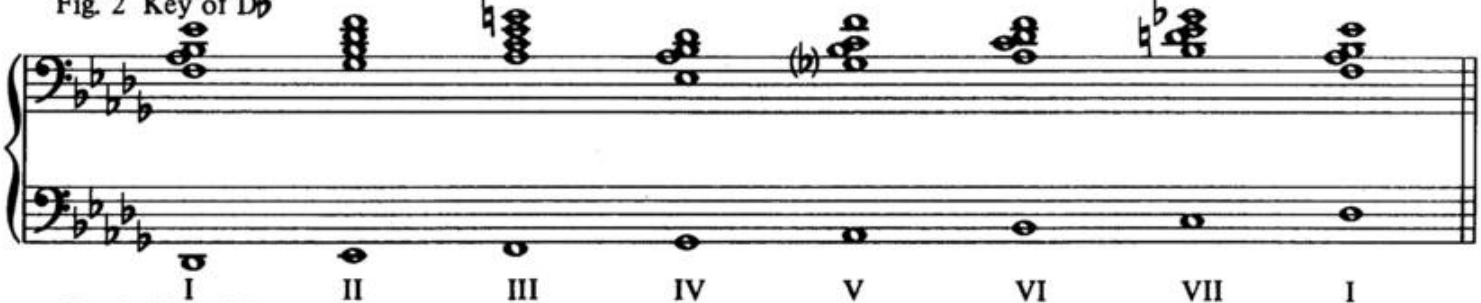


Fig. 3 Key of D

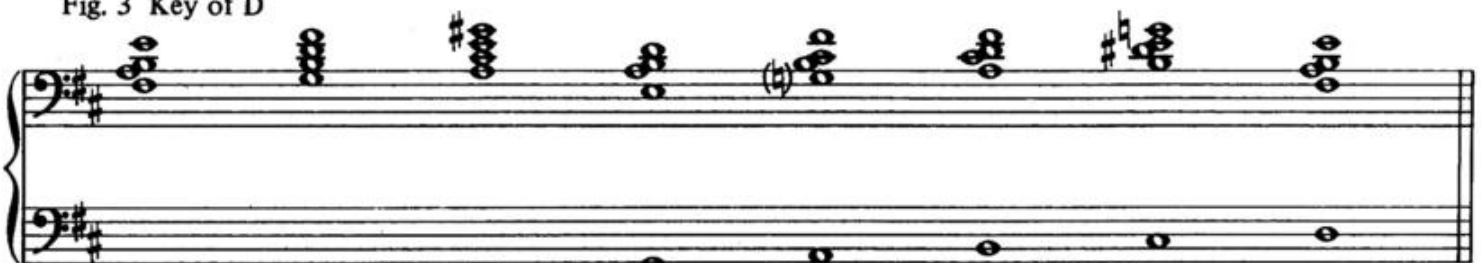


Fig. 4 Key of Eb

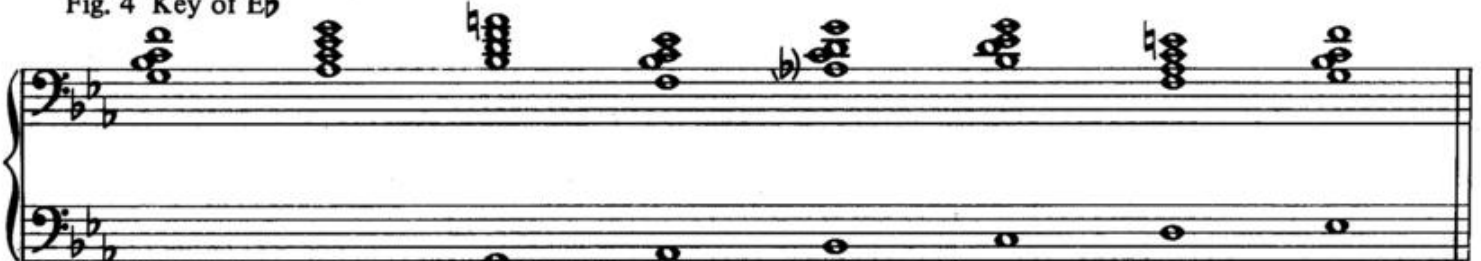


Fig. 5 Key of E



Fig. 6 Key of F

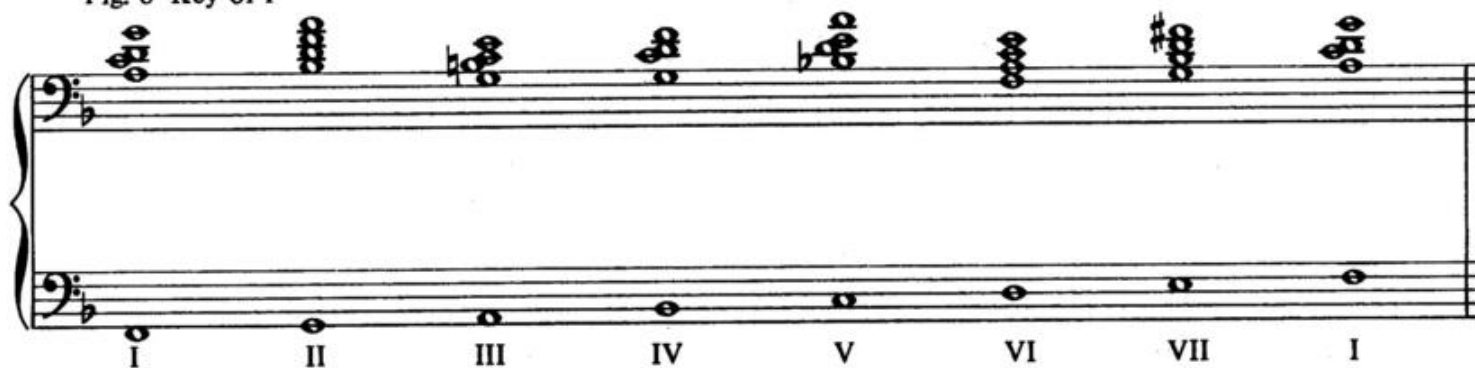


Fig. 7 Key of F $\sharp$
(Enharmonic of G $\flat$)

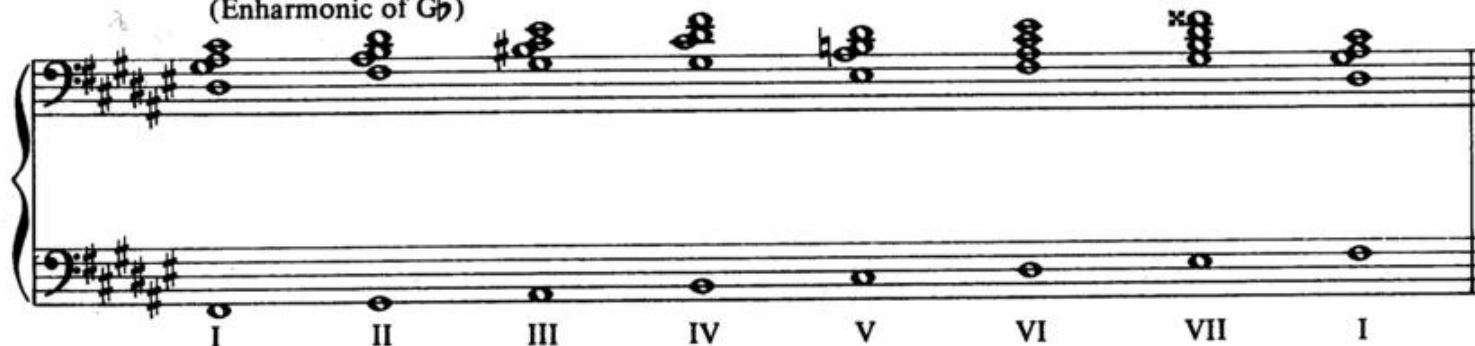


Fig. 8 Key of G $\flat$
(Enharmonic of F $\sharp$)

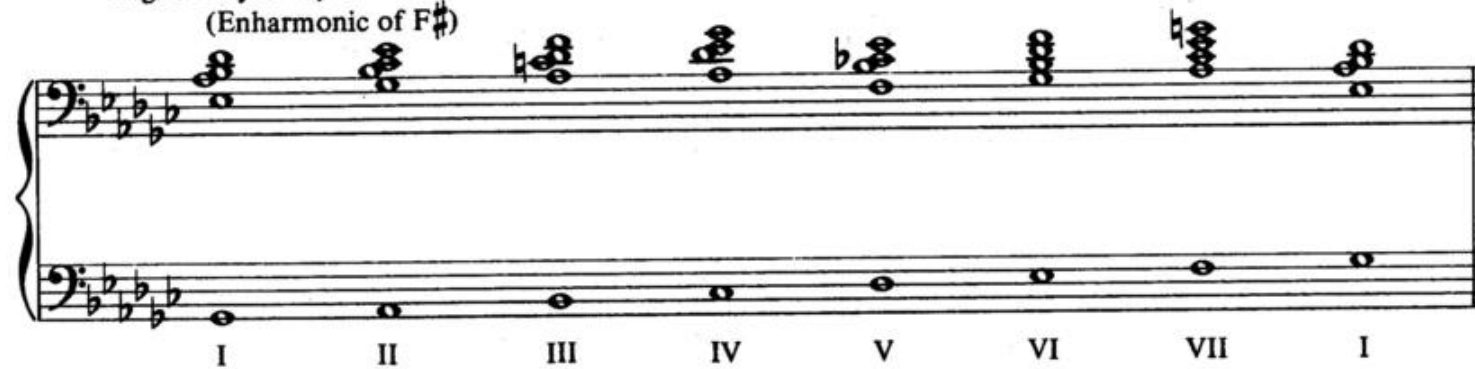


Fig. 9 Key of G

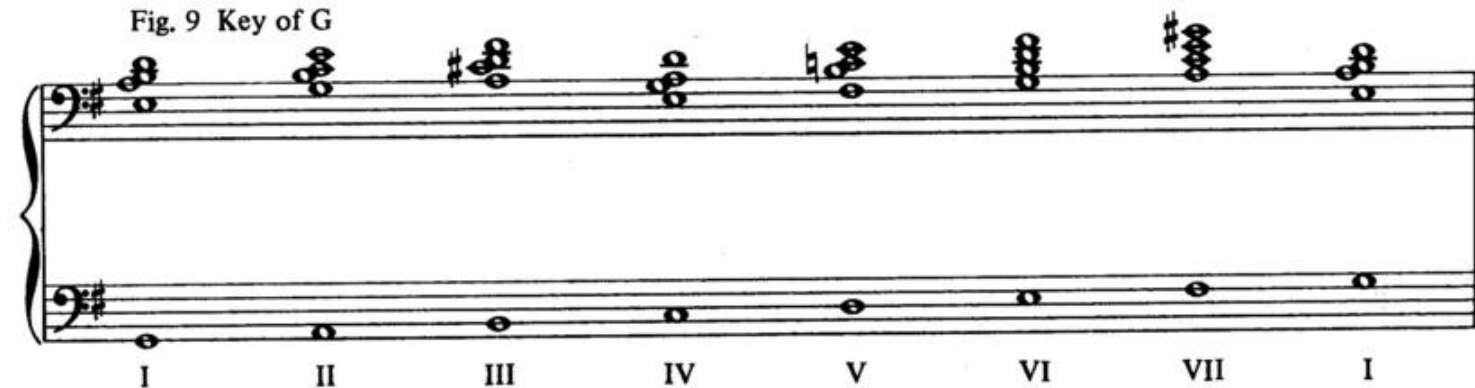


Fig. 10 Key of A $\flat$

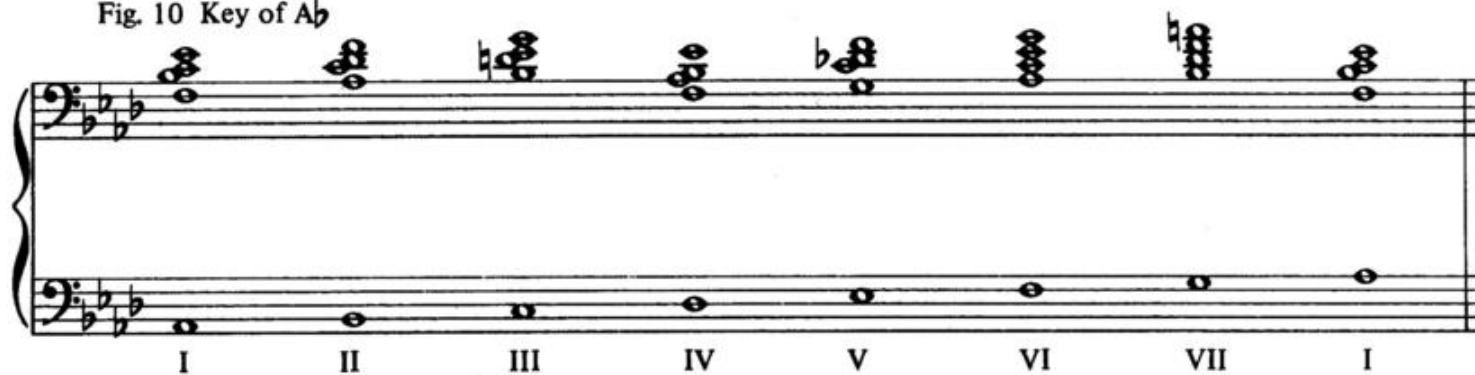


Fig. 11 Key of A



Fig. 12 Key of B $\flat$

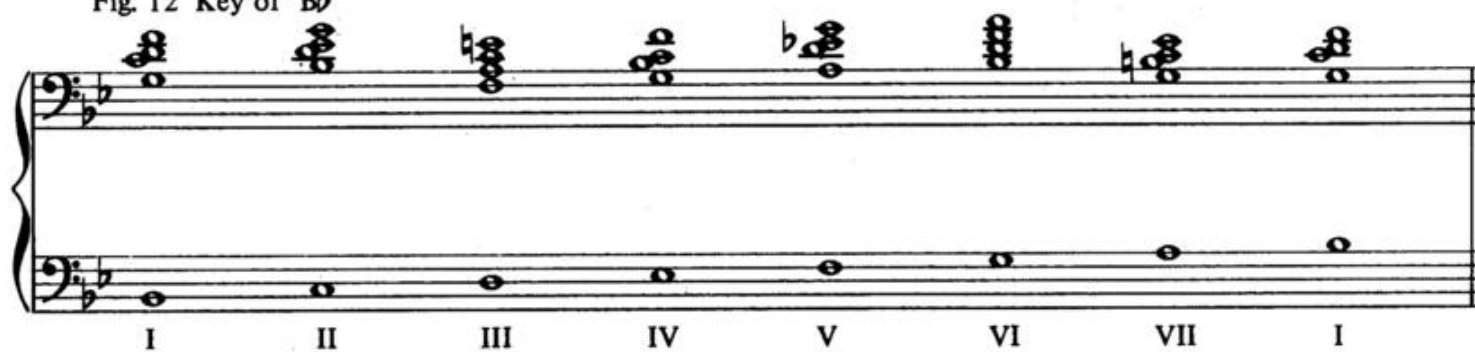


Fig. 13 Key of B



Chapter 22

Contemporary Left Hand Voicings Five Qualities

Figures 1 through 12 illustrate the voicing system applied to the five chord qualities.

Fig. 1

Figure 1 displays five voicings for the C major chord (CM) in the left hand. The voicings are shown on a grand staff (treble and bass clefs). The notes are: CM (C4, E4, G4), Cx (C4, E4, G4, Bb4), Cm (C4, E4, G4, Bb4, C5), Cø (C4, E4, G4, Bb4, C5, D5), and Co (C4, E4, G4, Bb4, C5, D5, E5). The notes are written in a compact, stacked format.

Fig. 2

Figure 2 displays five voicings for the D-flat major chord (DbM) in the left hand. The voicings are shown on a grand staff (treble and bass clefs). The notes are: DbM (Bb3, Db4, F4), Dbx (Bb3, Db4, F4, Ab4), C#m (Bb3, Db4, F4, Ab4, C5), C#ø (Bb3, Db4, F4, Ab4, C5, D5), and C#o (Bb3, Db4, F4, Ab4, C5, D5, E5). The notes are written in a compact, stacked format.

Fig. 3

Figure 3 displays five voicings for the D minor chord (Dm) in the left hand. The voicings are shown on a grand staff (treble and bass clefs). The notes are: DM (D4, F4, Ab4), Dx (D4, F4, Ab4, Bb4), Dm (D4, F4, Ab4, Bb4, C5), Dø (D4, F4, Ab4, Bb4, C5, D5), and Do (D4, F4, Ab4, Bb4, C5, D5, E5). The notes are written in a compact, stacked format.

Fig. 4

Fig. 4 displays five chords in the bass clef, each with its corresponding chord symbol below it:

- E_bM
- E_bx
- E_bm
- $E_b\emptyset$
- $E_b\circ$

Fig. 5

Fig. 5 displays five chords in the bass clef, each with its corresponding chord symbol below it:

- $E M$
- $E x$
- $E m$
- $E \emptyset$
- $E \circ$

Fig. 6

Fig. 6 displays five chords in the bass clef, each with its corresponding chord symbol below it:

- $F M$
- $F x$
- $F m$
- $F \emptyset$
- $F \circ$

Fig. 7

Fig. 7 displays five chords in the bass clef, each with its corresponding chord symbol below it:

- $F^\# M$
- $F^\# x$
- $F^\# m$
- $F^\# \emptyset$
- $F^\# \circ$

Fig. 8

GM Gx Gm Gø Go

Fig. 9

A♭M A♭x A♭m A♭ø A♭o

Fig. 10

AM Ax Am Aø Ao

Fig. 11

B♭M B♭x B♭m B♭ø B♭o

Fig. 12

BM Bx Bm Bø Bo

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